

claude-windows / e8c0dc7c-b2b1-4e19-91cc-46bfdb813e98

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\e8c0dc7c-b2b1-4e19-91cc-46bfdb813e98.jsonl`
- SHA-256: `2885af14b9322fd25580fb84fdf5777523da89b04b091ac5f1844181daff9d6e`
- Source modified: `2026-06-27T17:24:44+00:00`
- Imported at: `2026-07-05T16:48:19+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `e8c0dc7c-b2b1-4e19-91cc-46bfdb813e98`
```

Transcript

1. user (2026-06-07T07:47:10.822Z)

In sports-data-collector, `src/parsers/cvat/rally\_cvat.py::load\_canonical\_csv` (~lines 442-444) drops any rally whose start_time is exactly 0.0. The bug is the falsy-zero `or`:

```
start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
end = _parse_secs(row.get("end_time")) or _parse_secs(row.get("end_mmss"))
if start is None or end is None:
    continue
```

When `start\_time` parses to 0.0, `0.0 or \_parse\_secs(start\_mmss)` evaluates to the second operand (None if there's no start_mmss column), so the row is skipped. A rally that legitimately starts at 0.0s is silently lost (confirmed: a 3-row canonical CSV with row 1 at start_time=0 loads only 2 records).

Fix: prefer the decimal column, falling back to mm:ss only when the decimal is None, e.g.:

```
start = _parse_secs(row.get("start_time"))
if start is None:
    start = _parse_secs(row.get("start_mmss"))
end = _parse_secs(row.get("end_time"))
if end is None:
    end = _parse_secs(row.get("end_mmss"))
```

Add a regression test in tests/ (e.g. test_cvat_parser.py or a new test) that a canonical CSV whose first rally has start_time=0.0 (no start_mmss column) loads all rows. Run `python -m pytest -q` to confirm green.

2. assistant (2026-06-07T07:47:18.791Z)

I'll start by locating the `sports-data-collector` repo, since the bug is in a different project than my current working directory.

3. user (2026-06-07T07:47:22.851Z)

```
Name
----
.NET
.NET Aspire
Annotation Setup
badminton-highlight-indexer
badminton-highlight-indexer-khelacharya
CODE WITH ERIC
Email-Mining
gtest
muneem
rally-annotator
```

sports-data-collector
sports-obsreport
Verifiers

4. user (2026-06-07T07:47:25.663Z)

```
410         if ":" in s:
411             try:
412                 total = 0.0
413                 for part in s.split(":"):
414                     total = total * 60 + float(part)
415                 return total
416             except ValueError:
417                 return None
418         try:
419             return float(s)
420         except ValueError:
421             return None
422
423
424     def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
425         """Load a canonical per-rally CSV into RallyRecords, so a *CSV* delivery (not
426         just a CVAT export) can be validated. Decimal start/end preferred, mm:ss
427         fallback; rows with no usable start/end are skipped. (frame_* are -1 here, as a
428         CSV carries no frame info ? only the frame-derived checks become inapplicable.)
429         """
430         import csv as _csv
431         if not os.path.exists(csv_path):
432             raise FileNotFoundError(f"CSV not found: {csv_path}")
433         records: List[RallyRecord] = []
434         with open(csv_path, newline="", encoding="utf-8") as f:
435             reader = _csv.DictReader(f)
436             reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or
437 [])]
438             for raw in reader:
439                 row = {(k or "").strip().lower(): v for k, v in raw.items()}
440                 rn = safe_int(row.get("rally_number"), None)
441                 if rn is None:
442                     continue
443                 start = _parse_secs(row.get("start_time")) or
444 _parse_secs(row.get("start_mmss"))
445                 end = _parse_secs(row.get("end_time")) or _parse_secs(row.get("end_mmss"))
446                 if start is None or end is None:
447                     continue
448                 records.append(RallyRecord(
449                     rally_number=rn, start_time=start, end_time=end,
450                     shots_count=safe_int(row.get("shots_count"), None),
451                     ending_reason=(row.get("ending_reason") or "").strip() or None,
452                     last_shot_outcome=(row.get("last_shot_outcome") or "").strip() or None,
453                     score_before=(row.get("score_before") or "").strip() or None,
454                     score_after=(row.get("score_after") or "").strip() or None,
455                     notes=(row.get("notes") or "").strip() or None,
456                     frame_min=-1, frame_max=-1, n_boxes=0,
457                 ))
458             return records
459
460     def build_summary(records: List[RallyRecord], issues: List[Issue],
461                       source: str = "", fps: Optional[float] = None) -> Dict[str, Any]:
462         """Machine-readable validation summary (JSON / CI)."""
463         by_sev = {s: [i for i in issues if i.severity == s]
464                   for s in ("blocker", "high", "medium", "low")}
465         flagged = sorted({i.rally_number for i in issues if i.rally_number is not None})
466         return {
467             "source": source,
```

```

467         "fps": fps,
468         "n_rallies": len(records),
469         "n_issues": len(issues),
470         "counts_by_severity": {s: len(v) for s, v in by_sev.items()},
471         "n_blockers": len(by_sev["blocker"]),
472         "passed": len(by_sev["blocker"]) == 0,
473         "flagged_rallies": flagged,
474         "issues": [{"severity": i.severity, "code": i.code,
475                     "rally_number": i.rally_number, "message": i.message} for i in
issues],
476     }
477
478
479     def render_validation_md(records: List[RallyRecord], issues: List[Issue], source: str)
-> str:

```

5. assistant (2026-06-07T07:47:28.996Z)

Confirmed the bug at lines 442-443. Now let me apply the fix.

6. user (2026-06-07T07:47:33.594Z)

The file C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py has been updated successfully. (file state is current in your context ? no need to Read it back)

7. assistant (2026-06-07T07:47:38.068Z)

Now let me find the test directory and any existing CVAT parser tests.

8. user (2026-06-07T07:47:39.167Z)

```

C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py
C:\Users\avidu\Projects\sports-data-
collector\tests\__pycache__\test_cvat_parser.cpython-313-pytest-9.0.3.pyc

```

9. user (2026-06-07T07:47:39.559Z)

```

C:\Users\avidu\Projects\sports-data-collector\tests\test_downloader.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_local_import.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_schemas.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_validator.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_cli.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_config.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_parser.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_parsing.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_db.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py
C:\Users\avidu\Projects\sports-data-collector\tests\test_reporting.py

```

10. assistant (2026-06-07T07:47:42.054Z)

There's an existing `test\_cvat\_parser.py`. Let me read it to match style and find the right place for a regression test.

11. user (2026-06-07T07:47:42.229Z)

```

1     """Tests for the CVAT rally adapter (src/parsers/cvat/rally_cvat.py).
2
3     The adapter's whole reason for existing is to (a) recompute rally time from the
4     box FRAME RANGE at the true fps ? ignoring the vendor's wrong start/end_time
5     attributes ? and (b) report content defects rather than silently fixing them.
6     These tests pin both behaviours, plus the end-to-end hand-off to import-local.
7     """
8     import csv as _csv

```

```

9
10 from src.parsers.cvat.rally_cvat import (
11     RallyRecord, parse_cvat_boxes, build_records, detect_issues, convert,
12     write_canonical_csv, load_canonical_csv, build_summary, render_validation_md,
13     write_review_csv,
14 )
15 from src.cli.curate import SportsDataCLI
16
17
18 # --- helpers -----
19
20 def _img_box(frame, attrs):
21     rows = "".join(f'<attribute name="{k}">{v}</attribute>' for k, v in attrs.items())
22     return (f'<image id="{frame}" name="frame_{frame:06d}" width="1920" height="1080">'
23           f'<box label="rally" xtl="1" ytl="1" xbr="2" ybr="2">{rows}</box></image>')
24
25
26 def _write_cvat(path, images, step=30, stop_frame=1000):
27     body = "".join(images)
28     xml = (
29         '<?xml version="1.0" encoding="utf-8"?><annotations><version>1.1</version>'
30         f'<meta><job><stop_frame>{stop_frame}</stop_frame>'
31         f'<frame_filter>step={step}</frame_filter></job></meta>'
32         f'{body}</annotations>'
33     )
34     path.write_text(xml, encoding="utf-8")
35     return str(path)
36
37
38 def _rec(rn, start, end, shots=5, reason="winner", fmin=None, fmax=None, n=3,
outcome=None):
39     return RallyRecord(
40         rally_number=rn, start_time=start, end_time=end, shots_count=shots,
41         ending_reason=reason, last_shot_outcome=outcome,
42         score_before=None, score_after=None, notes=None,
43         frame_min=fmin if fmin is not None else int(start * 60),
44         frame_max=fmax if fmax is not None else int(end * 60), n_boxes=n)
45
46
47 # --- the core fix: time from frames, vendor times ignored -----
48
49 def test_time_recomputed_from_frames_not_attributes(tmp_path):
50     # rally 1 spans source frames 60..120; the vendor's start/end attributes are
51     # deliberately wrong (the frame/900 bug). At 60 fps the truth is 1.0..2.0s.
52     xml = _write_cvat(tmp_path / "j.xml", [
53         _img_box(60, {"rally_number": "1.0", "start_time": "0.2", "end_time": "1.4",
54                      "shots_count": "6.0", "ending_reason": "winner",
55                      "score_before (A-B)": "0-0", "notes": "clean"}),
56         _img_box(90, {"rally_number": "1.0", "start_time": "0.2", "end_time": "1.4"}),
57         _img_box(120, {"rally_number": "1.0", "start_time": "0.2", "end_time": "1.4"}),
58     ])
59     res = convert(xml, fps=60.0)
60     assert len(res.records) == 1
61     r = res.records[0]
62     assert r.start_time == 1.0 and r.end_time == 2.0           # frames/fps, not 0.2/1.4
63     assert r.vendor_start_time == "0.2"                       # preserved for transparency
64     assert r.shots_count == 6
65     assert r.ending_reason == "winner"
66     assert r.score_before == "0-0"                            # "(A-B)" alias resolved
67     assert r.notes == "clean"
68
69
70 def test_step_and_fps_read_from_meta_and_arg(tmp_path):
71     xml = _write_cvat(tmp_path / "j.xml", [
72         _img_box(0, {"rally_number": "1", "shots_count": "3"}),

```

```

73     _img_box(30, {"rally_number": "1"}),
74 ], step=30)
75 res = convert(xml, fps=60.0)
76 assert res.step == 30
77 assert res.fps == 60.0
78
79
80 def test_fps_required_when_unavailable(tmp_path):
81     xml = _write_cvat(tmp_path / "j.xml", [_img_box(0, {"rally_number": "1"})])
82     try:
83         convert(xml) # no fps, no video_path
84     except ValueError as e:
85         assert "fps" in str(e).lower()
86     else:
87         raise AssertionError("expected ValueError when fps cannot be determined")
88
89
90 # --- track-mode parsing + outside filtering -----
91
92 def test_track_mode_and_outside_filtering(tmp_path):
93     xml = (
94         '<?xml version="1.0"?><annotations><meta><job>'
95         '<stop_frame>1000</stop_frame><frame_filter>step=30</frame_filter></job></meta>'
96         '<track id="0" label="rally">'
97         '<box frame="60" outside="0" xtl="1" ytl="1" xbr="2" ybr="2">'
98         '<attribute name="rally_number">1</attribute>'
99         '<attribute name="shots_count">4</attribute></box>'
100         '<box frame="120" outside="0" xtl="1" ytl="1" xbr="2" ybr="2">'
101         '<attribute name="rally_number">1</attribute></box>'
102         '<box frame="150" outside="1" xtl="1" ytl="1" xbr="2" ybr="2">' # terminator
103         '<attribute name="rally_number">1</attribute></box>'
104         '</track></annotations>'
105     )
106     p = tmp_path / "track.xml"
107     p.write_text(xml, encoding="utf-8")
108     meta, boxes = parse_cvat_boxes(str(p))
109     assert len(boxes) == 2 # the outside=1 terminator dropped
110     recs = build_records(boxes, fps=60.0)
111     assert recs[0].frame_max == 120 # not 150 (the terminator)
112     assert recs[0].end_time == 2.0
113
114
115 # --- issue detection: surfaced, not silently fixed -----
116
117 def test_overlap_is_a_blocker():
118     recs = [_rec(1, 1.0, 5.0), _rec(2, 4.0, 8.0)] # r2 starts before r1 ends
119     issues = detect_issues(recs)
120     overlaps = [i for i in issues if i.code == "overlap"]
121     assert len(overlaps) == 1 and overlaps[0].severity == "blocker"
122
123
124 def test_zero_duration_is_a_blocker():
125     recs = [_rec(1, 2.0, 2.0, fmin=120, fmax=120, n=1)]
126     issues = detect_issues(recs)
127     assert any(i.code == "zero_duration" and i.severity == "blocker" for i in issues)
128
129
130 def test_offvocab_ending_reason_flagged():
131     recs = [_rec(1, 1.0, 3.0, reason="into_net")]
132     issues = detect_issues(recs)
133     assert any(i.code == "ending_reason_vocab" for i in issues)
134     # the adopted forced/unforced terms are accepted, not flagged
135     ok = detect_issues(_rec(1, 1.0, 3.0, reason="forced_error"))
136     assert not any(i.code == "ending_reason_vocab" for i in ok)
137

```

```

138
139 def test_low_shot_density_flags_inflated_span():
140     recs = [_rec(1, 0.0, 16.0, shots=2)]          # 2 shots over 16s -> 0.12/s
141     issues = detect_issues(recs)
142     assert any(i.code == "shot_density" for i in issues)
143
144
145 def test_outcome_vocab_flagged():
146     bad = detect_issues([_rec(1, 1.0, 3.0, outcome="long")])
147     assert any(i.code == "outcome_vocab" for i in bad)
148     # the agreed outcomes are accepted, not flagged
149     ok = detect_issues([_rec(1, 1.0, 3.0, outcome="out")])
150     assert not any(i.code == "outcome_vocab" for i in ok)
151
152
153 # --- end-to-end: emitted CSV imports via the existing import-local parser ---
154
155 def test_canonical_csv_round_trips_through_import_local(tmp_path):
156     xml = _write_cvat(tmp_path / "j.xml", [
157         _img_box(60, {"rally_number": "1", "start_time": "0.2", "end_time": "1.4",
158             "shots_count": "6", "ending_reason": "winner",
159             "last_shot_outcome": "in"}),
160         _img_box(120, {"rally_number": "1"}),
161         _img_box(180, {"rally_number": "2", "shots_count": "4",
162             "ending_reason": "forced_error", "last_shot_outcome": "out"}),
163         _img_box(240, {"rally_number": "2"}),
164     ])
165     res = convert(xml, fps=60.0)
166     assert res.n_blockers == 0
167     assert res.records[1].last_shot_outcome == "out"    # parsed from the box attribute
168     out = tmp_path / "canonical.csv"
169     write_canonical_csv(res.records, str(out))
170
171     # Re-read with the actual import-local parser (self is unused -> None is safe).
172     items = SportsDataCLI._parse_local_annotations(None, "badminton", "vid1", str(out))
173     assert len(items) == 2
174     assert items[0].start_time == 1.0 and items[0].end_time == 2.0    # not 0.2/1.4
175     assert items[1].ending_reason == "forced_error"
176     assert items[1].last_shot_outcome == "out"          # survives the round-trip into
the schema
177     assert items[1].start_time == 3.0 and items[1].end_time == 4.0
178
179
180 # --- validate-delivery: gap heuristic, CSV loading, report artifacts -----
181
182 def test_possible_missed_rally_gap_flagged():
183     # 20s of unlabeled time between two rallies -> flag (catches the 1:46-style miss)
184     recs = [_rec(1, 95.6, 98.0), _rec(2, 118.0, 119.6)]
185     issues = detect_issues(recs)
186     assert any(i.code == "possible_missed_rally" for i in issues)
187
188
189 def test_normal_gap_not_flagged():
190     recs = [_rec(1, 10.0, 14.0), _rec(2, 22.0, 26.0)]    # 8s gap -> fine
191     issues = detect_issues(recs)
192     assert not any(i.code == "possible_missed_rally" for i in issues)
193
194
195 def test_load_canonical_csv_decimal_and_mmss(tmp_path):
196     p = tmp_path / "c.csv"
197     p.write_text(
198         "rally_number,start_time,end_time,start_mmss,end_mmss,shots_count,ending_reason,
last_shot_outcome\n"
199         "1,3.0,21.0,,15,unforced_error,out\n"
200         "2,,0:29.000,0:32.000,2,winner,in\n", encoding="utf-8")

```

```

201     recs = load_canonical_csv(str(p))
202     assert len(recs) == 2
203     assert recs[0].start_time == 3.0 and recs[0].last_shot_outcome == "out"
204     assert recs[1].start_time == 29.0 and recs[1].end_time == 32.0    # mm:ss fallback
205
206
207     def test_validation_artifacts(tmp_path):
208         recs = [_rec(1, 1.0, 5.0), _rec(2, 4.0, 8.0, reason="into_net")]    # overlap + bad
vocab
209         issues = detect_issues(recs)
210         summary = build_summary(recs, issues, source="vid", fps=59.94)
211         assert summary["n_rallies"] == 2
212         assert summary["passed"] is False and summary["n_blockers"] >= 1
213         assert 2 in summary["flagged_rallies"]
214         md = render_validation_md(recs, issues, "vid")
215         assert "BLOCKED" in md and "## BLOCKER" in md
216         out = tmp_path / "review.csv"
217         write_review_csv(recs, issues, str(out))
218         rows = list(_csv.DictReader(open(out, encoding="utf-8")))
219         assert "VERIFIED(Y/N)" in rows[0] and "auto_flags" in rows[0]
220         assert any("overlap" in r["auto_flags"] for r in rows)
221

```

12. assistant (2026-06-07T07:47:47.658Z)

The existing test file has a perfect spot ? right after
`test\_load\_canonical\_csv\_decimal\_and\_mmss`. Let me add the regression test that pins the zero-start bug.

13. user (2026-06-07T07:47:53.275Z)

The file C:\Users\avidu\Projects\sports-data-collector\tests\test_cvat_parser.py has been updated successfully. (file state is current in your context ? no need to Read it back)

14. assistant (2026-06-07T07:48:01.493Z)

Now let me run the test suite in the `sports-data-collector` repo to confirm green.

15. user (2026-06-07T07:48:05.418Z)

```

..... [ 63%]
..... [100%]
113 passed in 1.76s

```

16. assistant (2026-06-07T07:48:09.958Z)

All 113 pass. Let me confirm the new regression test is meaningful ? that it actually fails against the old buggy logic ? by running just that test against a temporary revert.

17. user (2026-06-07T07:48:15.285Z)

```

. [100%]
1 passed in 0.18s
OLD start for start_time=0.0: None -> row dropped

```

18. assistant (2026-06-07T07:48:21.109Z)

The fix is complete and verified. Here's a summary:

What was wrong

In [rally_cvat.py:442](C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py:442), `load\_canonical\_csv` used a falsy-`or` fallback:
```python

```
start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
...
```

When `start\_time` parsed to `0.0`, `0.0 or ?` is falsy, so Python returned the second operand (`None` when there's no `start\_mmss` column), tripping `if start is None: continue` ? silently dropping any rally that legitimately starts at `0.0s`. The demo above confirms it: `OLD start for start\_time=0.0: None -> row dropped`.

## The fix

Replaced both `or` fallbacks with explicit `is None` checks, so the decimal column is preferred and mm:ss is used only when the decimal is genuinely absent:

### 19. user (2026-06-07T07:48:21.630Z)

```
442 start = _parse_secs(row.get("start_time"))
443 if start is None:
444 start = _parse_secs(row.get("start_mmss"))
445 end = _parse_secs(row.get("end_time"))
446 if end is None:
447 end = _parse_secs(row.get("end_mmss"))
448 if start is None or end is None:
449 continue
```

### 20. assistant (2026-06-07T07:48:26.691Z)

## Regression test

Added `test\_load\_canonical\_csv\_keeps\_rally\_starting\_at\_zero` in [tests/test\_cvat\_parser.py](C:\Users\avidu\Projects\sports-data-collector\tests\test\_cvat\_parser.py) ? a 3-row canonical CSV (no `start\_mmss` column) whose first rally has `start\_time=0.0`, asserting all 3 rows load and `recs[0].start\_time == 0.0`.

## Verification

- `python -m pytest -q` ? **\*\*113 passed\*\*** (was 112 + the new test).
- Confirmed the new test is a genuine guard: the old falsy-`or` logic returns `None` for `start\_time=0.0`, which would have dropped the row and failed the `len(recs) == 3` assertion.

These are edits to the **``sports-data-collector``** repo (a sibling of your current **``badminton-highlight-indexer``** working dir), not committed ? let me know if you'd like me to branch and commit there.

### 21. user (2026-06-07T07:49:16.023Z)

please branch and commit and make sure its a part of the PR

### 22. assistant (2026-06-07T07:49:20.868Z)

Let me check the git state of the **``sports-data-collector``** repo ? its current branch, status, and any open PRs ? to understand what "part of the PR" means here.

### 23. user (2026-06-07T07:49:25.890Z)

```
=== current branch ===
feat/observability-reports
=== status ===
=== recent commits ===
de5c0ff feat(reporting): per-video observability report (collector)
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
2882a03 feat(ingest): validate-delivery ? per-video QA gate + structured report
23f2ed0 Merge pull request #8 from avidullu/feat/last-shot-outcome
810541e feat(schema): add last_shot_outcome (in/net/out/n_a) alongside ending_reason
30d1d73 Merge pull request #7 from avidullu/refactor/ending-reason-forced-unforced
fe764d6 Merge pull request #6 from avidullu/feat/cvat-rally-adapter
aae93c0 fix(ingest): drop coarse_sampling issue from CVAT adapter
=== open PRs ===
```



=== PR for current branch (if any) ===

Relevant pull requests in avidullu/sports-data-collector

Current branch

There is no pull request associated with [feat/observability-reports]

Created by you

You have no open pull requests

Requesting a code review from you

You have no pull requests to review

#### 24. assistant (2026-06-07T07:49:36.553Z)

The working tree shows clean, but I edited two files there ? let me double-check git is seeing the right tree and where the repo root is.

#### 25. user (2026-06-07T07:49:39.989Z)

Exit code 1

=== git rev-parse --show-toplevel ===

C:/Users/avidu/Projects/sports-data-collector

=== git status (full) ===

On branch feat/observability-reports

nothing to commit, working tree clean

=== diff stat ===

=== check-ignore on the two files ===

#### 26. user (2026-06-07T07:49:48.059Z)

```
437 for raw in reader:
438 row = {(k or "").strip().lower(): v for k, v in raw.items()}
439 rn = safe_int(row.get("rally_number"), None)
440 if rn is None:
441 continue
442 start = _parse_secs(row.get("start_time"))
443 if start is None:
444 start = _parse_secs(row.get("start_mmss"))
445 end = _parse_secs(row.get("end_time"))
446 if end is None:
447 end = _parse_secs(row.get("end_mmss"))
448 if start is None or end is None:
449 continue
450 records.append(RallyRecord(
451 rally_number=rn, start_time=start, end_time=end,
```

#### 27. assistant (2026-06-07T07:49:54.698Z)

The file on disk has my fix, yet git reports a clean tree ? meaning HEAD already contains this exact content. Let me check what HEAD has for these lines and the file's commit history.

#### 28. user (2026-06-07T07:49:59.123Z)

=== HEAD version of the fix region ===

```
start = _parse_secs(row.get("start_time"))
if start is None:
 start = _parse_secs(row.get("start_mmss"))
end = _parse_secs(row.get("end_time"))
if end is None:
 end = _parse_secs(row.get("end_mmss"))
if start is None or end is None:
 continue
```

=== HEAD version of the new test (grep) ===

```
def test_load_canonical_csv_keeps_rally_starting_at_zero(tmp_path):
=== last commits touching these files ===
de5c0ff feat(reporting): per-video observability report (collector)
2882a03 feat(ingest): validate-delivery ? per-video QA gate + structured report
810541e feat(schema): add last_shot_outcome (in/net/out/n_a) alongside ending_reason
aae93c0 fix(ingest): drop coarse_sampling issue from CVAT adapter
be75d80 feat(ingest): CVAT rally export -> canonical CSV adapter
=== rlog (last 5) ===
de5c0ff HEAD@{0}: commit: feat(reporting): per-video observability report (collector)
d68c5da HEAD@{1}: checkout: moving from master to feat/observability-reports
d68c5da HEAD@{2}: pull --ff-only: Fast-forward
23f2ed0 HEAD@{3}: checkout: moving from feat/validate-delivery to master
2882a03 HEAD@{4}: commit: feat(ingest): validate-delivery ? per-video QA gate + structured
report
```

## 29. assistant (2026-06-07T07:50:35.680Z)

Both my fix **\*\*and\*\*** my new test are already inside commit `de5c0ff` ("feat(reporting): per-video observability report"). That's unexpected ? in my last turn I told you the edits were uncommitted. I need to understand exactly what happened before I branch, so I don't fabricate a story. Let me inspect that commit.

## 30. user (2026-06-07T07:50:39.735Z)

```
=== de5c0ff metadata ===
hash=de5c0ff12c93556b1dca88bffb45767bdec0fb11
author=Avi Dullu
date=2026-06-07 13:18:41 +0530
subject=feat(reporting): per-video observability report (collector)
=== files changed in de5c0ff ===
 data/roora_badminton_canonical.csv | 25 +++
 data/roora_badminton_issues.txt | 16 ++
 data/sports_metadata.db | Bin 184320 -> 200704 bytes
 data/validation/pilot_badminton_01_report.json | 45 ++++++
 data/validation/pilot_badminton_01_report.md | 16 ++
 data/validation/pilot_badminton_01_review.csv | 25 +++
 docs/INGESTION_PIPELINE.md | 22 +++
 nonexistent.report.json | 89 ++++++
 nonexistent.report.md | 42 +++++
 nonexistent.summary.md | 28 +++++
 pyproject.toml | 11 +-
 src/cli/curate.py | 38 +++++
 src/parsers/cvat/rally_cvat.py | 8 +-
 src/reporting/__init__.py | 68 ++++++
 src/reporting/adapters.py | 151 ++++++
 src/reporting/spec.yaml | 47 ++++++
 tests/test_cvat_parser.py | 15 ++
 tests/test_reporting.py | 120 ++++++
18 files changed, 763 insertions(+), 3 deletions(-)
=== does de5c0ff touch rally_cvat.py / test_cvat_parser.py? (diff excerpt) ===
commit de5c0ff12c93556b1dca88bffb45767bdec0fb11
Author: Avi Dullu <avi.dullu@gmail.com>
Date: Sun Jun 7 13:18:41 2026 +0530
```

```
feat(reporting): per-video observability report (collector)
```

Emits <id>.report.json/.report.md (technical) + <id>.summary.md (customer/vendor delivery summary) via the shared obsreport library (SOFT dependency; never raises).

- src/reporting/: soft-import wrapper + adapter (rally\_cvat Issue -> Flag; delivery + import recorders reusing existing records/issues) + spec.yaml (sections, customer\_verdict, license\_noncommercial + missing\_fps rules).
- curate.py: wire validate-delivery (alongside existing \_report.\* artifacts), convert-cvat (next to canonical CSV), import-local (next to video). Each cites docs/INGESTION\_PIPELINE.md; vendor-facing summary shows plain issue text only.

- docs/INGESTION\_PIPELINE.md: FAQ for the report + non-commercial-export note.
- Fix build-backend typo (setuptools.build\_meta) so pip install -e works.
- 9 new tests. Suite 113 green; verified end-to-end via validate-delivery CLI smoke.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

```
diff --git a/src/parsers/cvat/rally_cvat.py b/src/parsers/cvat/rally_cvat.py
index 6b9c595..6b18266 100644
--- a/src/parsers/cvat/rally_cvat.py
+++ b/src/parsers/cvat/rally_cvat.py
@@ -439,8 +439,12 @@ def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
 rn = safe_int(row.get("rally_number"), None)
 if rn is None:
 continue
- start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
- end = _parse_secs(row.get("end_time")) or _parse_secs(row.get("end_mmss"))
+ start = _parse_secs(row.get("start_time"))
+ if start is None:
+ start = _parse_secs(row.get("start_mmss"))
+ end = _parse_secs(row.get("end_time"))
+ if end is None:
+ end = _parse_secs(row.get("end_mmss"))
 if start is None or end is None:
 continue
 records.append(RallyRecord(
diff --git a/tests/test_cvat_parser.py b/tests/test_cvat_parser.py
index 4c241f8..11fb310 100644
--- a/tests/test_cvat_parser.py
+++ b/tests/test_cvat_parser.py
@@ -204,6 +204,21 @@ def test_load_canonical_csv_decimal_and_mmss(tmp_path):
 assert recs[1].start_time == 29.0 and recs[1].end_time == 32.0 # mm:ss fallback

+def test_load_canonical_csv_keeps_rally_starting_at_zero(tmp_path):
+ # Regression: a rally that legitimately starts at 0.0s must not be dropped.
+ # The old `start_time or start_mmss` fallback treated the falsy 0.0 as "missing"
+ # and skipped the row when there was no start_mmss column to fall back to.
+ p = tmp_path / "c.csv"
+ p.write_text(
+ "rally_number,start_time,end_time,shots_count,ending_reason,last_shot_outcome\n"
+ "1,0.0,4.0,8,winner,in\n"
+ "2,10.0,14.0,6,forced_error,out\n"
+ "3,20.0,24.0,5,unforced_error,out\n", encoding="utf-8")
+ recs = load_canonical_csv(str(p))
+ assert len(recs) == 3 # the zero-start row 1 is not lost
```

### 31. assistant (2026-06-07T07:51:22.647Z)

Now I have the full picture. Let me verify the push/remote state so I can present accurate options.

### 32. user (2026-06-07T07:51:27.260Z)

```
=== branches (local + remote) ===
* feat/observability-reports
 master
 remotes/origin/feat/cvat-rally-adapter
 remotes/origin/feat/last-shot-outcome
 remotes/origin/feat/validate-delivery
 remotes/origin/master
 remotes/origin/refactor/ending-reason-forced-unforced
=== upstream tracking for current branch ===
fatal: no upstream configured for branch 'feat/observability-reports'
=== commits on feat/observability-reports not on master ===
d4f9c15 feat(reporting): per-video observability report (collector)
```

=== is de5c0ff on origin anywhere? ===

### 33. assistant (2026-06-07T07:51:37.661Z)

The commit hash changed (`de5c0ff` ? `d4f9c15`) between my last two checks ? the repo is being modified concurrently (likely you, or a parallel session). Let me re-read the current state before doing anything.

### 34. user (2026-06-07T07:51:42.527Z)

```
=== current HEAD ===
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
=== working tree status ===
M .gitignore
M data/sports_metadata.db
M docs/INGESTION_PIPELINE.md
M pyproject.toml
M src/cli/curate.py
M src/parsers/cvat/rally_cvat.py
M tests/test_cvat_parser.py
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
?? data/validation/
?? src/reporting/
?? tests/test_reporting.py
=== is my fix in current HEAD? ===
 start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
 end = _parse_secs(row.get("end_time")) or _parse_secs(row.get("end_mmss"))
 if start is None or end is None:
=== is my test in current HEAD? ===
=== reflog 3 ===
d68c5da HEAD@{0}: reset: moving to HEAD~1
d4f9c15 HEAD@{1}: commit (amend): feat(reporting): per-video observability report (collector)
de5c0ff HEAD@{2}: commit: feat(reporting): per-video observability report (collector)
```

### 35. assistant (2026-06-07T07:52:14.825Z)

The reflog now tells the full story:

- `de5c0ff` ? original reporting commit (which had swept up my fix)
- `d4f9c15` ? that commit **amended**
- `d68c5da` ? then **git reset HEAD~1** undid it

So someone (you, or a parallel session) just un-committed the reporting work. HEAD is back at the master tip (`d68c5da`), and everything ? the reporting feature **and** my zero-start fix ? is now sitting uncommitted in the working tree again. Let me confirm exactly what's in the working tree for my two fix files.

### 36. user (2026-06-07T07:52:18.223Z)

```
=== working-tree diff: rally_cvat.py (my fix) ===
diff --git a/src/parsers/cvat/rally_cvat.py b/src/parsers/cvat/rally_cvat.py
index 6b9c595..6b18266 100644
--- a/src/parsers/cvat/rally_cvat.py
+++ b/src/parsers/cvat/rally_cvat.py
@@ -439,8 +439,12 @@ def load_canonical_csv(csv_path: str) -> List[RallyRecord]:
 rn = safe_int(row.get("rally_number"), None)
 if rn is None:
 continue
- start = _parse_secs(row.get("start_time")) or _parse_secs(row.get("start_mmss"))
- end = _parse_secs(row.get("end_time")) or _parse_secs(row.get("end_mmss"))
+ start = _parse_secs(row.get("start_time"))
+ if start is None:
+ start = _parse_secs(row.get("start_mmss"))
```

```

+ end = _parse_secs(row.get("end_time"))
+ if end is None:
+ end = _parse_secs(row.get("end_mmss"))
+ if start is None or end is None:
+ continue
+ records.append(RallyRecord(
=== working-tree diff: test_cvat_parser.py (my test) ===
diff --git a/tests/test_cvat_parser.py b/tests/test_cvat_parser.py
index 4c241f8..11fb310 100644
--- a/tests/test_cvat_parser.py
+++ b/tests/test_cvat_parser.py
@@ -204,6 +204,21 @@ def test_load_canonical_csv_decimal_and_mmss(tmp_path):
 assert recs[1].start_time == 29.0 and recs[1].end_time == 32.0 # mm:ss fallback

+def test_load_canonical_csv_keeps_rally_starting_at_zero(tmp_path):
+ # Regression: a rally that legitimately starts at 0.0s must not be dropped.
+ # The old `start_time` or `start_mmss` fallback treated the falsy 0.0 as "missing"
+ # and skipped the row when there was no start_mmss column to fall back to.
+ p = tmp_path / "c.csv"
+ p.write_text(
+ "rally_number,start_time,end_time,shots_count,ending_reason,last_shot_outcome\n"
+ "1,0.0,4.0,8,winner,in\n"
+ "2,10.0,14.0,6,forced_error,out\n"
+ "3,20.0,24.0,5,unforced_error,out\n", encoding="utf-8")
+ recs = load_canonical_csv(str(p))
+ assert len(recs) == 3 # the zero-start row 1 is not lost
+ assert recs[0].start_time == 0.0 and recs[0].end_time == 4.0
+
+def test_validation_artifacts(tmp_path):
+ recs = [_rec(1, 1.0, 5.0), _rec(2, 4.0, 8.0, reason="into_net")] # overlap + bad vocab
+ issues = detect_issues(recs)

```

### 37. assistant (2026-06-07T07:52:43.571Z)

The working tree cleanly isolates my fix to exactly two files (`rally\_cvat.py` + `test\_cvat\_parser.py`), separate from the reporting feature files. So I can commit it cleanly.

Two things I want to confirm before I touch git, because the situation has changed since your instruction:

1. **\*\*The repo is being actively rewritten\*\*** ? within the last few minutes the reporting commit was created, amended, then undone with `git reset HEAD~1`. If a parallel session or you are mid-operation, my branching could collide.
2. **\*\*"The PR" is ambiguous now.\*\*** There is no open PR in `sports-data-collector` yet, and the reporting work is currently uncommitted alongside my fix.

Given the bug fix is logically independent of the reporting feature, I want to confirm where it should land.

### 38. user (2026-06-07T07:52:47.260Z)

The user did not answer the questions.

### 39. assistant (2026-06-07T07:53:24.176Z)

I'll proceed with the recommended approach ? a dedicated branch + its own PR for the fix (the cleanest reading of "branch and commit and make it part of the PR," and it leaves the in-progress reporting work untouched). Let me first re-verify the repo state is stable (no rebase/merge mid-flight) before I touch git.

### 40. user (2026-06-07T07:53:29.090Z)

```

=== HEAD now ===
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
=== master ref ===
d68c5da
=== in-progress op? (rebase/merge dirs) ===
(none above = clean)
=== remote ===
origin https://github.com/avidullu/sports-data-collector.git (fetch)
origin https://github.com/avidullu/sports-data-collector.git (push)

```

#### 41. assistant (2026-06-07T07:53:40.932Z)

HEAD changed again ? there's now a commit `**`028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv`**`. The fix is being committed concurrently (this is almost certainly you working in parallel, which is why the question went unanswered). I'll stop mutating and just read the current state so I don't create a duplicate.

#### 42. user (2026-06-07T07:53:45.728Z)

```

=== current branch ===
feat/observability-reports
=== log -4 ===
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
bb2b5be feat(reporting): per-video observability report (collector)
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
2882a03 feat(ingest): validate-delivery ? per-video QA gate + structured report
=== 028937b files ===
 src/parsers/cvat/rally_cvat.py | 8 ++++++--
 tests/test_cvat_parser.py | 15 ++++++
 2 files changed, 21 insertions(+), 2 deletions(-)
=== working tree ===
M data/sports_metadata.db
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
?? data/validation/
=== local + remote branches ===
* feat/observability-reports
 master
 remotes/origin/feat/cvat-rally-adapter
 remotes/origin/feat/last-shot-outcome
 remotes/origin/feat/validate-delivery
 remotes/origin/master
 remotes/origin/refactor/ending-reason-forced-unforced
=== which remote branches contain 028937b ===
=== open PRs ===

```

#### 43. assistant (2026-06-07T07:54:39.379Z)

The repo has stabilized, and the concurrent work (you, in parallel) has already produced exactly the clean structure I was aiming for. Here's the `**current, verified state**` of ``sports-data-collector``:

```

Branch `feat/observability-reports` (local only ? not pushed, no PR yet):
...
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv ? the fix
bb2b5be feat(reporting): per-video observability report (collector) ? reporting
feature
d68c5da Merge pull request #9 ... ? master tip
...

```

- The zero-start fix is now its `**own clean commit**` (``028937b``), touching only the two files ? ``src/parsers/cvat/rally_cvat.py`` (+the ``is None`` fix) and ``tests/test_cvat_parser.py`` (+the regression test). Diff is byte-identical to what I verified earlier at `**113 tests green**`.
- It's been cleanly `**separated**` from the reporting feature (``bb2b5be``).

- Working tree now only has data artifacts (`sports\_metadata.db`, untracked `data/...` outputs)  
? no code changes outstanding.

So "branch and commit" is effectively **done**. The only remaining piece is "make it part of the PR" ? and there's **no PR yet** and the branch isn't pushed.

I didn't push/open a PR on my own because (a) it's an outward-facing action and (b) the branch currently bundles the reporting commit too, so a PR for this branch would contain **both** commits. How would you like to finish it?

#### 44. user (2026-06-07T08:03:45.441Z)

Your questions have been answered: "How should I get the fix into a PR? (Branch feat/observability-reports has both the fix commit and the reporting commit, not yet pushed.)"="Fix on its own branch + PR". You can now continue with these answers in mind.

#### 45. assistant (2026-06-07T08:04:14.014Z)

You chose **fix on its own branch + PR**. Let me execute: re-verify state, create `fix/canonical-csv-zero-start` off master, and cherry-pick the fix commit onto it.

#### 46. user (2026-06-07T08:04:18.764Z)

```
=== re-verify HEAD + that bb2b5be does NOT touch the fix files ===
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
(empty above = reporting commit doesn't touch the fix files -> clean cherry-pick)
=== create fix branch off master ===
M data/sports_metadata.db
Switched to a new branch 'fix/canonical-csv-zero-start'
=== cherry-pick the fix commit ===
[fix/canonical-csv-zero-start f7cab11] fix(parser): keep a rally that starts at 0.0s in
load_canonical_csv
Date: Sun Jun 7 13:22:53 2026 +0530
2 files changed, 21 insertions(+), 2 deletions(-)
=== verify ===
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
M data/sports_metadata.db
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
?? data/validation/
```

#### 47. assistant (2026-06-07T08:04:33.561Z)

Clean standalone branch created: `fix/canonical-csv-zero-start` = master + the single fix commit (`f7cab11`). Let me run the test suite on this isolated branch to confirm green before pushing.

#### 48. user (2026-06-07T08:04:37.105Z)

```
..... [68%]
..... [100%]
105 passed in 1.65s
```

#### 49. assistant (2026-06-07T08:04:41.747Z)

105 passed (the 8-test difference is just the reporting tests, which live on the other branch). The fix + regression test are green in isolation. Now I'll push the branch and open the PR against master.

#### 50. user (2026-06-07T08:04:46.057Z)

```
=== push ===
branch 'fix/canonical-csv-zero-start' set up to track 'origin/fix/canonical-csv-zero-start'.
remote:
```

remote: Create a pull request for 'fix/canonical-csv-zero-start' on GitHub by visiting:

```
remote: https://github.com/avidullu/sports-data-collector/pull/new/fix/canonical-
csv-zero-start
remote:
To https://github.com/avidullu/sports-data-collector.git
 * [new branch] fix/canonical-csv-zero-start -> fix/canonical-csv-zero-start
```

#### 51. assistant (2026-06-07T08:04:54.082Z)

Branch pushed. Now I'll open the PR against master.

#### 52. user (2026-06-07T08:04:59.192Z)

<https://github.com/avidullu/sports-data-collector/pull/10>

#### 53. assistant (2026-06-07T08:05:06.765Z)

PR opened: `**[sports-data-collector#10](https://github.com/avidullu/sports-data-collector/pull/10)**`. Now let me switch the working tree back to ``feat/observability-reports`` so you're left where you were, and confirm the final state.

#### 54. user (2026-06-07T08:05:11.223Z)

```
M data/sports_metadata.db
Switched to branch 'feat/observability-reports'
=== final: current branch + log ===
feat/observability-reports
028937b fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
bb2b5be feat(reporting): per-video observability report (collector)
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
=== fix branch tracking ===
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
d68c5da Merge pull request #9 from avidullu/feat/validate-delivery
=== PR ===
{"baseRefName":"master","headRefName":"fix/canonical-csv-zero-
start","number":10,"state":"OPEN","title":"fix(parser): keep a rally that starts at 0.0s in
load_canonical_csv","url":"https://github.com/avidullu/sports-data-collector/pull/10"}
```

#### 55. assistant (2026-06-07T08:05:20.932Z)

Done. The zero-start fix now lives on its own branch with a standalone PR.

### Summary

```
[sports-data-collector#10](https://github.com/avidullu/sports-data-collector/pull/10) ?
`fix(parser): keep a rally that starts at 0.0s in load_canonical_csv` (OPEN, `fix/canonical-csv-
zero-start` ? `master`)
```

- **Branch:** ``fix/canonical-csv-zero-start`` = ``master`` + one commit (``f7cab11``), pushed and tracking origin.
- **Contents:** only the two fix files ? the ``is None`` fallback fix in ``src/parsers/cvat/rally_cvat.py`` and the ``test_load_canonical_csv_keeps_rally_starting_at_zero`` regression test. No reporting work bundled in.
- **Verified:** ``python -m pytest -q`` ? **105 passed** on the isolated branch.

### A note on what happened mid-task

While I was deliberating, the repo was being rewritten in parallel (the reporting commit got created, amended, reset, then re-split into two clean commits ? ``bb2b5be`` reporting + ``028937b`` the fix). I cherry-picked that fix commit onto the new branch rather than re-committing my working-tree edits, so the PR matches the commit you'd already made.



I left the working tree back on `**`feat/observability-reports`**`, exactly where you were. That branch still has both commits locally (the reporting feature ``bb2b5be`` and the fix ``028937b``) and is `**not pushed / has no PR**` ? untouched, for you to handle separately. The only working-tree changes are unrelated data artifacts (``data/sports_metadata.db`` and untracked ``data/...`` outputs), which I did not commit.

#### 56. user (2026-06-07T08:09:40.673Z)

PR is merged.

#### 57. assistant (2026-06-07T08:10:24.245Z)

PR merged ? that's a checkpoint. Let me verify the merge, do the post-merge cleanup, and update the session memory. First, gather state (git/PR) and read the checkpoint files in parallel.

#### 58. user (2026-06-07T08:10:30.918Z)

```
=== PR #10 merge state ===
{"baseRefName":"master","headRefName":"fix/canonical-csv-zero-start","mergeCommit":{"oid":"99dfa22cbab1a40aa63a154105219b7adad562f8"},"mergedAt":"2026-06-07T08:08:51Z","mergedBy":{"id":"MDQ6VXNlcjE2NzQ4Nzc=","is_bot":false,"login":"avidullu","name":"Avi"},"number":10,"state":"MERGED"}
=== fetch + prune ===
From https://github.com/avidullu/sports-data-collector
- [deleted] (none) -> origin/feat/cvat-rally-adapter
- [deleted] (none) -> origin/feat/last-shot-outcome
- [deleted] (none) -> origin/feat/validate-delivery
- [deleted] (none) -> origin/fix/canonical-csv-zero-start
- [deleted] (none) -> origin/refactor/ending-reason-forced-unforced
d68c5da..99dfa22 master -> origin/master
=== remote branches matching fix/ ===
=== local branches ===
* feat/observability-reports
 fix/canonical-csv-zero-start
 master
=== current branch + working tree ===
feat/observability-reports
?? data/roora_badminton_canonical.csv
?? data/roora_badminton_issues.txt
?? data/validation/
```

#### 59. assistant (2026-06-07T08:10:48.557Z)

PR #10 is `**MERGED**` (merge commit ``99dfa22``, merged by avidullu), and GitHub auto-deleted the remote ``fix/canonical-csv-zero-start`` branch (the fetch also pruned several other already-merged branches). Let me sync local master, clean up the local fix branch, and read the checkpoint files to update them.

#### 60. user (2026-06-07T08:10:55.113Z)

```
=== fast-forward local master to origin/master (not checked out) ===
From https://github.com/avidullu/sports-data-collector
d68c5da..99dfa22 master -> master
=== is the fix commit f7cab11 an ancestor of master? (0 = yes, safe to -d) ===
exit=0
=== delete merged local fix branch ===
error: the branch 'fix/canonical-csv-zero-start' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D fix/canonical-csv-zero-start'
hint: Disable this message with "git config advice.forceDeleteBranch false"
=== confirm fix is in master ===
 start = _parse_secs(row.get("start_time"))
 if start is None:
 start = _parse_secs(row.get("start_mmss"))
```

```
=== final local branches ===
* feat/observability-reports bb2b5be [origin/feat/observability-reports] feat(reporting): per-
video observability report (collector)
 fix/canonical-csv-zero-start f7cab11 [origin/fix/canonical-csv-zero-start: gone] fix(parser):
keep a rally that starts at 0.0s in load_canonical_csv
 master 99dfa22 [origin/master] Merge pull request #10 from
avidullu/fix/canonical-csv-zero-start
```

#### 61. assistant (2026-06-07T08:11:08.152Z)

The fix is confirmed in master (merge commit `99dfa22` = "Merge pull request #10", and `f7cab11` is an ancestor of master ? `is-ancestor` exit 0). The `-d` refusal is just because `git` checks against the current branch (`feat/observability-reports`), which holds the cherry-pick `source` `028937b`, not `f7cab11` ? the patch is definitely in master, so `-D` is safe. I also notice `feat/observability-reports` now has an upstream (`origin/feat/observability-reports`) ? you've pushed the reporting branch too.

Let me delete the merged local fix branch and read the checkpoint files to update them.

#### 62. user (2026-06-07T08:11:12.542Z)

```
Exit code 1
Deleted branch fix/canonical-csv-zero-start (was f7cab11).
=== remaining branches ===
* feat/observability-reports bb2b5be [origin/feat/observability-reports] feat(reporting): per-
video observability report (collector)
 master 99dfa22 [origin/master] Merge pull request #10 from
avidullu/fix/canonical-csv-zero-start
=== is the reporting branch's fix commit (028937b) going to dedupe vs master on rebase? check
patch-id presence ===
028937b in master? exit=1 (1=no, will dedupe by patch on rebase)
```

#### 63. assistant (2026-06-07T08:11:35.732Z)

Branch cleanup done. Final git state is clean:

- **master** = `99dfa22` ? includes the zero-start fix (PR #10).
- **feat/observability-reports** = `bb2b5be` (reporting only, now pushed to `origin/feat/observability-reports`) ? note the redundant `028937b` fix commit has been dropped from this branch (by your concurrent cleanup), so the reporting branch no longer duplicates what's in master.

Now the checkpoint update. PR-merged is a trigger per my working agreement ? let me read the live checkpoint files to update them minimally (without duplicating git history).

#### 64. user (2026-06-07T08:11:37.119Z)

```
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-07 (later) . ? **NEW FEATURE IN PROGRESS ? per-video
OBSERVABILITY REPORTS** across
13 BOTH tools (indexer + collector) + a NEW shared lib. Approved plan:
14 `C:\Users\avidu\claudel\plans\drifting-snuggling-squirrel.md` (READ IT to resume).
Decisions: always-on,
15 reports written NEXT TO THE VIDEO by default (override via `report.output_dir`); BOTH
```

```

tools emit a technical
16 report (debuggable by a README+FAQ reader) AND a customer-friendly summary (strip
internals); shared schema via
17 a **new private repo `sports-obsreport`** (pkg `obsreport`) ? a reusable "report config
language."
18 **Research:** Soda/SodaCL = ELv2 (disqualified, not OSS);
GE/Evidently/ydata/Frictionless license-safe but
19 wrong shape ? built a small lib on permissive primitives (pydantic/Jinja2/PyYAML).
20 - ? **PHASE C DONE ? `sports-obsreport` lib built + LOCAL git tag `v0.1.0`** at
`C:\Users\avidu\Projects\sports-obsreport\`
21 (NOT on GitHub yet ? local only; editable-installed into Python 3.13 via `pip install
-e . --no-build-isolation`).
22 71 tests green. Modules: `envelope.py` (typed ReportEnvelope = single source of
truth), `spec.py` (YAML config
23 language: sections + rules), `rules.py` (sub-Turing clause engine, no eval),
`render/{json_render,markdown}.py`
24 (technical + customer views from same envelope), `io.py` (next-to-video paths, atomic,
NEVER raises),
25 `recorder.py` (RunRecorder: in-flight stage capture, finalize?rules?verdict, write),
`diff.py` (stable_json for
26 cross-run diffs), `schema/report_envelope.schema.json` (contract + drift test).
specs/example.yaml ships
27 silent_provider_noop + fps_fallback_used. **Footgun caught:** correct build-backend is
`setuptools.build_meta`
28 (underscore) ? collector's pyproject has the hyphen typo `setuptools.build-meta` =
latent `pip install -e` break (flag/fix).
29 - ? **obsreport bumped to v0.1.1** (local tag) ? added optional `customer_message` on
flags/rules: customer report
30 shows ONLY curated flags (no technical leakage); technical always uses `message`.
Schema still 1.0 (additive).
31 - ? **PHASE A DONE (indexer)** on branch `feat/observability-reports` (NOT pushed;
commit 9dbc764). 222 tests green
32 (209 baseline + 13). Added `run_all_with_context` (backward-compat; run_all
unchanged); `backend/reporting/`
33 (soft-import wrapper ? no-op if obsreport absent + never-raises; pure `harvest.py`;
`spec.yaml` footgun rules);
34 wired `main.py` CLI + API in finally blocks; README FAQ Q10-Q13; requirements.txt
commented git+ssh dep. Verified
35 end-to-end via CLI smoke (report files appear next to video; customer summary clean;
technical findings cite FAQ).
36 - ? **PHASE B DONE (collector)** on branch `feat/observability-reports` (NOT pushed).
113 tests green. `src/reporting/`
37 (soft-import wrapper ? never raises; `adapter.py` Issue?Flag + delivery/import
recorders reusing rally_cvat;
38 `spec.yaml` sections + customer_verdict + item_noun=rally +
license_noncommercial/missing_fps rules). Wired
39 curate.py validate-delivery / convert-cvat / import-local; import report falls back to
annotations dir when video
40 absent (not cwd). docs/INGESTION_PIPELINE.md FAQ added. Fixed build-backend typo.
.gitignore ignores
41 *.report.json/.report.md/.summary.md. Verified via validate-delivery CLI smoke.
42 - ?? **obsreport bumped to v0.1.3** during B (local tags v0.1.0/.1/.2/.3): added
customer_message (curated customer
43 flags), customer_verdict (per-tool verdict phrasing), item_noun
(rally/point/highlight). 74 lib tests green.
44 - ?? **DUP RISK:** the collector `load_canonical_csv` falsy-zero bug (drops a rally
starting at 0.0s) was fixed BOTH
45 as a separate commit on feat/observability-reports (`fix(parser)`, cherry-pickable)
AND by a user-started spawned
46 session (chip task_7a3d8081). **At PR time, keep only ONE** (recommend the spawned
session's dedicated PR; drop/skip
47 my fix commit). Don't double-merge.
48 - ? **SYNCED TO REMOTE (2026-06-07):** obsreport ? github.com/avidullu/sports-obsreport
(PRIVATE, main + tags
49 v0.1.0?v0.1.3). Indexer ? **PR #41** (rebased onto #39 typed-config `backend/config/`;

```

```

main.py conflict resolved
50 = typed attr access for segmenter + my reporting; reporting normalizes
AppConfig?dict). Collector ? **PR #11**
51 (observability-ONLY; I DROPPED my dup parser-fix commit ? Grok's `fix/canonical-csv-
zero-start` held branch owns it).
52 - ?? **Indexer PR #41 has 2 PRE-EXISTING red tests from #39** (NOT mine, left untouched
to avoid clashing with Grok's
53 config work): test_config.py::test_load_default_config (config.json
default_segmenter=gemini vs model default
54 yolo_hybrid) + test_save_default_config_roundtrip (Path==str on Windows). My
reporting: 226 pass. Collector: 112 pass.
55 - ?? **NEXT: finish PHASE D debuggability eval** (ultracode Workflow: fresh-reader
agents over broken-run reports ?
56 refine wording; render a report to an image + eyeball) + committed golden tests
(faq_ref-resolution + flag-set) in both repos.
57 - **TODO (HELD for owner OK):** create private GitHub repo `sports-obsreport` + push
(tags v0.1.0/v0.1.1); then
58 pin obsreport via git+ssh in both repos' deps; then push indexer/collector branches +
open PRs.
59 - Phasing tasks tracked in the harness task list (C1-C4 + A done; B in progress; D
pending).
60
61 --- (prior checkpoint) ---
62 **Checkpoint:** 2026-06-07 . ? **SESSION SUNSET.** master clean (PRs #25?#35 merged; 209
tests green; only
63 feat/khelacharya-shot-intelligence retained; no open PRs). Audio E1 done ? classical
onset = ~2.2x ceiling, NOT
64 adopted (detector kept, PR #35); quantitative findings + lessons in
[[audio-e1-findings]] (+ plan $10). VLC
65 annotator = standalone repo github.com/avidullu/rally-annotator (MIT, multi-sport).
66 ***? RESUME NEXT SESSION on EITHER:** (1) **GOLDEN EVALS** ? owner self-rates clips in
the VLC annotator ?
67 `import-local`/`--labels` ? unblocks distilled-model LOVO (n?3) + human-gold E1 re-
confirm = the highest-leverage
68 unblock; OR (2) **AUDIO E2** ? learned timbre classifier (needs labeled hits). Also
queued: pose/stance #2, owner's
69 "other hunch", Option B (owned end-to-end model). Read [[audio-e1-findings]], [[code-
map-artifact]] (docs/CODE_MAP.md),
70 docs/RALLY_DETECTION_RESEARCH.md, ADR-007 to ramp.
71 ? **PR #36 (docs/AUDIO_E1_FINDINGS.md) MERGED to master 2026-06-07** (owner restored the
branch after an accidental
72 close; reopened + squash-merged). Record now in-repo + [[audio-e1-findings]] (memory) +
plan $10. master fully clean, no open PRs.
73
74 --- (prior checkpoint) CLEAN SLATE. Indexer PRs #25?#32 MERGED; master 202 tests green,
tree clean, branches pruned
75 (kept feat/khelacharya-shot-intelligence). Audio decided (ADR-007). VLC annotator ?
standalone public repo
76 github.com/avidullu/rally-annotator (MIT, multi-sport). **ALL PRs #25?#34 MERGED; slate
FULLY clean** (master only + retained feat/khelacharya-shot-intelligence; no open
77 PRs; tree clean). #33 removed the stale annotator copy; #34 expanded ADR-007 (fusion +
Option B) + repointed the
78 ref ? both merged safely. **PARKED: modeling waits on golden labels (user self-rates via
annotator); next code
79 lever = AUDIO (E1 go/no-go probe).**
80 **ARCHITECTURE DECISION captured (ADR-007 $Integration architecture, PR #34):** audio
fuses in OUR rally layer
81 (WASB stays frozen black box ? feature-level into distill_local + decision-level recall-
recovery); **Option B = the
82 intended end-state** ? once a healthy golden corpus exists, train an OWNED end-to-end
multimodal model (frames+audio+
83 pose+??rallies) fine-tuned on our data, dropping WASB-weight dependence + recall
ceiling. Modular cues now (audio?
84 stance?other) are how we earn Option B (each generates the labels it needs). See
resumption runbook ?.

```

```

85
86 ## DISTILLED MODEL BUILT (2026-06-06) ? user wants this approach; honest result = needs
more data
87 - **ALL PRs #27-#31 MERGED to master** (0c6f2c1). master now has: code-map,
gate-A/rally_gate, gemini-refine
88 (+provider hardening + extract_video_chunk scale_width), calibrate_local,
distill_local. This session's 5
89 branches PRUNED (local+remote). Retained: feat/khelacharya-shot-intelligence (PR #11).
Pre-existing merged
90 stragglers still on origin (left, optional prune): docs/ending-reason-forced-
unforced(#19), feat/rally-slicer(#20),
91 feat/wasb-substrate(#21).
92 - **User will RATE the clips ? cached artifacts LEFT IN PLACE** (do NOT delete):
Temp\{adarsh,testlarge}_frames +
93 trajectories (~clips + Temp), Temp\verylarge_frames (partial 12246/46080).
Adarsh+TestLargeVideo trajectories
94 ready ? when user import-locals golden labels, calibrate_local/distill_local use them
with zero re-extraction.
95 (#30 stranded calibrate_local on gate branch ? RECOVERED in PR #31.)
96 - `backend/eval/distill_local.py` (PR #31, OPEN): learned model ? WASB recall-oriented
candidates (0.005,1.0,0.5)
97 ? features [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]
(resolution/fps-INVARIANT)
98 ? classifier.LogisticRegression trained on Gemini labels (training.label_candidates
IoU). Honest LOVO + saves
99 output/local_rally_model.json. Reuses
classifier/training/rally_gate/metrics/gemini_refine.
100 - **HONEST RESULT (2 videos): LEARNED UNDERPERFORMS baseline** ? mean held-out F1 0.359
(learned) vs 0.438
101 (threshold baseline). Causes: (1) n=2 too few ? no transferable model; (2) WASB
proposal recall CEILING 0.43
102 on TestLargeVideo (its windows don't align with Gemini rallies; Adarsh ceiling 0.81).
Framework READY; needs
103 more teacher videos + better WASB recall to beat baseline. Final weights:
mean_velocity + duration dominate,
104 net_crossings barely used (unreliable at n=2).
105 - ? **VeryLargeTestVideo (4K, 46080 frames, 12m49s, 11.5GB) = held-out FINAL test for
USER MANUAL EVAL.**
106 Extraction running (task bzt8hc4ks, frames?1920). Then WASB ? baseline + learned rally
lists ? user manually
107 evals (manual = GT; no Gemini needed on it). User's manual eval becomes a 3rd-video
human GT ? strengthens model.
108 - PR #31 (feat/distill-local) OPEN ? merge to get distill_local + calibrate_local onto
master.
109
110 ## ? AUDIO SCAFFOLDING + PRELIMINARY E1 DONE (2026-06-07) ? PR #35 (feat/audio-hit-
detector-e1, OPEN). Verdict = AMBER.
111 - Built `backend/audio/hit_detector.py` (ffmpeg?numpy spectral-flux onset?adaptive peak;
NO scipy; 5 tests) +
112 `backend/eval/audio_e1_probe.py` (E1 diagnostic). 207 tests green. WASB untouched
(Option A).
113 - Ran preliminary E1 on EXISTING GoPro audio vs **Gemini SILVER** labels (no wait for
human labels): Adarsh (36
114 rallies) + TestLargeVideo (7).
115 - **RESULT (honest, AMBER):** ? audio fires in **100% of rallies** and **100% of WASB-
missed rallies have
116 hit-clusters** ? real RECALL potential (the bottleneck). ?? but **discrimination
weak**: 1.3x @k=2.5, peaks
117 **~2.8x @k=4**, then a CLIFF (k=6 ? ~0 hits); hit rate high (56/min @k=2.5 = many non-
shuttle transients:
118 footwork/voices/ambient); **audio-only F1 0.13-0.27 < WASB-only (0.33-0.54).** So
audio is NOT clean enough
119 to stand alone ? it's a **fusion/recall cue**, not a standalone segmenter.
120 - **OPTION A DONE (band-pass) ? NEGATIVE RESULT (2026-06-07, committed to PR #35):**
added a `band` knob to the

```

121 onset detector + swept low/mid/high bands × k at 32kHz on both videos. **\*\*Band-pass**  
does NOT help\*\* ? HF

122 "shuttle-click" bands gave LOWER discrimination, not higher. **\*\*Ceiling stays ~2.2x\*\***  
(in-rally vs MID-MATCH

123 dead-time). **\*\*Not a warm-up/label artifact\*\*** (mid-match dead alone ~0.6 hits/s vs ~1.3  
in-rally on both).

124 Root cause: between-point activity (shuttle tosses=our false-fire, footwork, talk) =  
transients a classical

125 onset detector can't separate from rally hits. `k` only trades coverage for  
discrimination. Kept the band knob

126 (cheap/optional) but it's NOT the fix. Full writeup: AUDIO\_RALLY\_DETECTION\_PLAN.md  
§10.

127 - **\*\*HONEST VERDICT: classical-onset audio is too weak to be a clean cue (~2.2x); band-**  
pass won't rescue it.\*\*

128 Remaining audio path = **\*\*E2 learned TIMBRE classifier\*\*** (MFCC ? shuttle-thwack vs  
toss/footstep/voice; needs

129 labeled hits ? real work, the only plausible way past 2.2x). Cheaper: re-confirm on  
human-gold (unlikely to

130 move much per the warm-up check). **\*\*Recommend re-prioritizing:\*\*** owner's "other hunch"  
/ pose-stance #2 /

131 more golden data for WASB+distilled, rather than over-investing in classical audio.  
**\*\*User to decide when back.\*\***

132 - **\*\*PR #35 review addressed (2026-06-07):\*\*** owner left 2 refactor comments (PR otherwise  
LGTM). (1) made

133 `backend/eval/audio/` package + moved probe ? `backend/eval/audio/e1\_probe.py` (folded  
INTO #35 since it's a

134 new unmerged file ? cleaner than a stacked PR; import/doc refs updated). (2) broader  
tests/ de-fragmentation

135 filed as a TODO in `docs/BACKLOG.md` (Code & test organization) ? grouping metric is  
owner's design call,

136 dedicated PR later. 209 tests green; replied on the PR. **\*\*#35 still OPEN, ready to  
merge as the single PR.\*\***

137

138 **## ? DECISION DOCUMENTED (2026-06-07):** AUDIO next, STANCE parked ? PR #32 (docs/audio-  
plan-adr007)

139 - **\*\*ADR-007\*\*** (docs/DECISIONS.md): adopt fully-local AUDIO "thwack" hit detection FUSED  
with WASB trajectory as

140 the next rally cue (attacks recall bottleneck); park pose/stance as #2 (harder);  
Gemini = offline-teacher only;

141 gate experiments on golden labels. **\*\*Sequence: AUDIO now ? [TBD owner hunch] ?  
POSE/STANCE.\*\***

142 - **\*\*docs/AUDIO\_RALLY\_DETECTION\_PLAN.md\*\*** ? detailed plan:  
extract?onset?peak?confidence?hit-cluster?fuse-with-WASB;

143 experiments E1(GO/NO-GO recall probe on real GoPro audio) / E2(LR+MFCC FP suppression)  
/ E3(audio features into

144 distill\_local) / E4(boundary tighten); deps numpy+scipy (permissive); top risk =  
shuttle SNR (quiet vs tennis).

145 New module planned: `backend/audio/hit\_detector.py`.

146 - **\*\*docs/VIDEO\_RECORDING\_GUIDELINES.md\*\*** ? AUDIO now a first-class capture REQUIREMENT  
(record w/ audio on, mic

147 unobstructed). User will add "enable audio always" to camera-setup docs.

148 - **\*\*Research checked in\*\*** (was uncommitted): docs/RALLY\_DETECTION\_RESEARCH.md. **\*\*VLC  
annotator checked in\*\***:

149 tools/vlc\_rally\_annotator/ (source+README). All in PR #32.

150 - Owner has "another hunch" to slot BEFORE stance (TBD).

151

152 **## ? RESEARCH DONE (2026-06-06) ? rally-detection methods ?**  
`docs/RALLY\_DETECTION\_RESEARCH.md` (full cited report)

153 Verdict: teacher?student is SOUND but NOT uniquely best ? augment with a COMPLEMENTARY  
LOCAL multi-cue stack.

154 **\*\*#1 cheapest lever = AUDIO hit detection (shuttle "thwack") fused with the trajectory ?  
directly attacks the**

155 WASB recall bottleneck with inputs we already have.\*\* #2 = pose/court serve-ready START  
+ flight-gradient END

156 (our "gate B", validated). #3 = compact HitNet GRU (MonoTrack ~16K params). Audio+visual

fusion lifts rally

157 RECALL (tennis HMM: 83% fused vs 54% visual/63% audio). Keep Gemini OFFLINE-teacher only. ? caveats: most lit

158 numbers are BROADCAST + per-frame (not IoU boundary F1, not amateur single-cam) ? won't transfer cleanly; ?

159 LICENSE TBD for ShuttleSet/MonoTrack/WASB-weights (commercial). Cheap next experiments E1-E4 in the doc (E1:

160 audio energy-peaks overlay on WASB windows ? recover misses; E3: add audio features to the distill student).

161

162 ## ? VLC RALLY ANNOTATOR ? STANDALONE PUBLIC REPO (2026-06-07):  
github.com/avidullu/rally-annotator

163 MOVED OUT of the indexer into its own \*\*MIT, public, multi-sport\*\* repo (default branch `main`) per user ? for

164 open-source release + iteration across \*\*net-separated racquet sports\*\* (badminton/tennis/table\_tennis/  
165 pickleball/padel). Local clone: `C:\Users\avidu\Projects\rally-annotator\` (git identity = Avi Dullu

166 <avi.dullu@gmail.com>, mirrored from indexer). Layout: `vlc/rally\_annotator.lua` (v1.2 ? added Sport dropdown +

167 `sport` CSV column; generic naming) + README + LICENSE(MIT) + docs/CSV\_FORMAT.md + CHANGELOG. CSV now

168 `rally\_number,start\_time,end\_time,ending\_reason,sport`. \*\*v1.2 INSTALLED\*\* to `%APPDATA%\vlc\lua\extensions\  
169 rally\_annotator.lua` (replaced v1.1). ? v1.2 multi-sport build NEEDS user's live VLC smoke test (I can't click

170 VLC). Removed from indexer (PR #32 now docs-only; ADR-007 ref repointed to the new repo).

171 Roadmap (in the new repo README): live test all 5 sports; per-sport hotkeys; python-vlc/HTML5 front-ends.

172 --- historical (v1.1, badminton-only, was in indexer tools/) ---

173 Button-dialog (not pause-hook;

174 pause callback flaky macOS/broken VLC4): View menu ? "Badminton Rally Annotator" ? Mark START / ending\_reason

175 dropdown (winner/forced\_error/unforced\_error/service\_fault/let/other) / Mark END / Undo + live status. Reads

176 vlc.var.get(input,"time")/1e6 ? decimal sec. Appends `rally\_number,start\_time,end\_time,ending\_reason` to

177 `

178 CSV ingests UNCHANGED by backend/eval/calibrate\_wasb.py --labels, backend/tools/rally\_slicer.py --labels, AND

179 collector import-local. Verify=high confidence (fixed dup-rally\_number, HTML status label, install-doc repo

180 mixup) but NEEDS user's live VLC test (I can't click VLC). Alternatives if cramped: python-vlc+Tkinter (global

181 hotkeys) or HTML5 page (shareable w/ remote raters). This is the golden-data generation tool ? feeds the model.

182

183 ## ? GOLDEN DATA via SELF-RATING (confirmed 2026-06-06) ? preferred over Gemini silver

184 User can be a RATER: make a CSV `rally\_number,start\_time,end\_time` (sec; JSON ok; optional shots\_count/ending\_reason)

185 ? `python -m src.cli.curate import-local --sport badminton --video-path <vid> --annotations-path <csv> --license Proprietary`

186 (validates rallies via validate\_badminton\_rallies, registers CURATED) ? `curate export` ? indexer eval format ?

187 add to calibrate\_local/distill\_local manifest. HUMAN golden > Gemini silver ? directly fixes the 2 caps (too few

188 videos + label quality). Collector `import\_local` @src/cli/curate.py:52; CSV parse @\_parse\_local\_annotations:144.

189 \*\*Pause decision 2026-06-06:\*\* modeling PAUSED until more golden data (user self-rating + next week's vendor videos).

190 VeryLargeTestVideo 4K extraction STOPPED at 12246/46080 (partial frames Temp\verylarge\_frames deletable). Research

191 workflow wxzp6h7xk still running ? post report then fully paused.

```

192
193 ## ?? NEXT-WEEK RESUMPTION RUNBOOK (more teacher videos ? re-run calibrate_local +
distill_local; learned beats baseline once n grows)
194 **Goal:** more teacher-labeled videos ? stable leave-one-video-out ? then build the
LEARNED distilled model
195 (classifier.py on Gemini labels) ? the real local-precision lever (threshold-tuning
ceiling'd at ~0.44 P/F1, overfits on 2 videos).
196 **FIRST: merge the open PR stack** so master has the tools: #27 (code-map), #28
(gate-A/rally_gate), #29
197 (gemini-refine + provider hardening + extract_video_chunk scale_width), #30
(calibrate_local, stacked on #28 ? retarget to master after #28).
198 **Per NEW video (repeat for each):**
199 1. Probe: `ffprobe ... <video>` ? note fps, width. (TestLargeVideo was 5.3K ? extract
frames downscaled to 1920.)
200 2. Extract frames: Windows `ffmpeg -i <video> [-vf scale=1920:-2] -q:v 2
<Temp>/<name>_frames/%08d.jpg`.
201 3. WASB trajectory (WSL, checkpointed): re-stage wasb_infer.py to `~/models/WASB-
SBDT/src`, then
202 `conda activate wasb && cd ~/models/WASB-SBDT/src && python wasb_infer.py
--frames_dir /mnt/c/.../<name>_frames
203 --weights ../pretrained_weights/wasb_badminton_best.pth.tar --out
~/clips/<name>_traj.csv --batch-size 8 --num-workers 4 --flush-every 500`
204 then copy CSV to a Windows path.
205 4. Gemini teacher labels (paid key in env GEMINI_API_KEY; NEVER store key): `python -m
backend.eval.gemini_refine
206 --video <video> --full-video --out output/<name>_gemini_fullvideo.csv` (model gemini-
flash-latest; clips auto-downscaled <500MB).
207 5. Add `{name,trajectory,fps,frame_width,labels}` to the calibration manifest (see
output/local_calib_manifest.json).
208 **Then re-run calibration + build the learned model:**
209 6. `python -m backend.eval.calibrate_local --manifest <manifest> --metric f1` (and
--metric precision) ? honest LOVO across N videos.
210 7. **NEW (the real lever):** train `backend/eval/classifier.py` (LogisticRegression) on
pooled Gemini labels: label each WASB
211 candidate rally-vs-not by IoU-match to Gemini labels (training.py::label_candidates),
features = candidate stats
212 (peak/mean velocity, active_frame_count) + net-crossings (rally_gate). Score held-out
(LOVO). Compare to threshold-calib.
213 **Artifacts that persist (don't redo):** trajectories
`~/clips/{adarsh,testlarge,sample_long}_traj.csv` (+ Windows Temp copies);
214 Gemini labels output/{testlarge,adarsh}_gemini_fullvideo.csv; manifest
output/local_calib_manifest.json.
215 **Deletable (big, regenerable from trajectories):** Temp/{adarsh_frames(7.9GB),
testlarge_frames, sample_long_frames(4.1GB)}.
216
217 ## OLD checkpoint context below ? (this session's journey)
218
219 ## DEMO RUN ? 2nd video "Adarsh and Vasanth.MP4" (2026-06-06, stress + checkpoint test)
220 - Video: `C:\Users\avidu\Projects\Annotation Setup\Collect\Adarsh and Vasanth.MP4`,
1920x1080, 59.94fps,
221 40,536 frames, 11m16s, 5.5GB. Stress-tested infra (1.9x Sample Long).
222 - Frames: ffmpeg -> `C:\?\Temp\adarsh_frames` (40,536 JPGs, 7.9GB, deletable). Inference
40,534 windows
223 in 1272s @ 31.9 win/s; 13,224/40,536 visible (32.6%). Checkpoint cache kept pace (no
crash, so resume
224 not exercised, but cache fully built). Trajectory: WSL `~/clips/adarsh_traj.csv` +
`C:\?\Temp\adarsh_traj.csv`.
225 - Tuned export (cfg 0.05/1.0/1.0, no golden labels for this video): **65 rally segments,
251.4s play** ->
226 `output/adarsh_tuned_segments.csv`. User eyeballing; baseline NOT yet generated (held
until user anchors).
227 NOTE tuned cfg was fit on Sample Long (different video) -> transfer is the open
question.
228
229 ## BASELINE vs TUNED on Adarsh (2026-06-06) ? tuned config does NOT transfer + over-fire

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```

pattern
230 - Baseline (0.02/2.0/2.0): 45 segs, 303.8s, mean 6.8s. Tuned (0.05/1.0/1.0): 65 segs,
251.4s, mean 3.9s.
231 - Tuned = +44% segments but -17% play: 30 tuned-only segs (almost all 1-2s = false
fires); 10 baseline-only
232 segs that tuned LOST are the LONG rallies (17.7/10.9/9.6/9.2/8.9/7.6s) ? merge_gap=1.0
fragments them.
233 - **Honest verdict: no "% improvement" (no GT for this video); tuned REGRESSES here.
Baseline is the better
234 operating point on Adarsh.** Concrete proof the Sample-Long-tuned cfg overfits -> need
leave_one_video_out
235 (2nd labelled video). `output/adarsh_baseline_segments.csv` written.
236 - **OVER-FIRE PATTERN (user-documented, data-confirmed):** loser tossing/flicking
shuttle back for service =
237 single short trajectory -> WASB detects as rally start. Real rally = shuttle crosses
net MULTIPLE times.
238 - **FIX SPECTRUM (design, grounded in current substrate):** (A) shuttle-only: gate
candidates on net/midline
239 crossings >=K (?shot count) / direction-reversals ? cheap, no new model, kills single-
toss fires. (B) user's
240 START-STANCE STATE MACHINE: IDLE->await-stance(players settled+diagonal in service
zones)->ARMED->shuttle->
241 RALLY->end->IDLE; reuse the existing person detector (yolo_hybrid torchvision
FRCNN+ByteTrack) as a VERIFIER
242 run only ~2s around each candidate start (not full-video, which was ~50min/video). (C)
pose-based serve
243 detection ? richest, new model + license review, likely overkill. Rec: A now, B robust
target; B needs research.
244 - ? **GATE A SHIPPED ? PR #28** (`feat/rally-start-gate`):
`backend/detectors/rally_gate.py` (pure,
245 camera-agnostic, no new model) ? estimate play axis (larger shuttle spread) + net
(median coord),
246 count (coord-net) sign-changes per window, keep crossings>=K. Wired into export via
`--min-crossings`
247 (0=off, 2 rec). 6 tests, suite 193 green. **Validated on Adarsh (side camera auto-
detected, net x?716):
248 real rallies 5-13 crossings, junk 0-1; K>=2 kills 18 of ~19 short tuned false fires,
keeps all long
249 rallies.** `output/adarsh_baseline_gated.csv` = BASELINE+gate(K=2) = 38 segs (45->38)
= the "stronger
250 baseline now". 1 yellow flag to eyeball: 4:39.5-4:44.7 (5.2s,1 crossing) removed ?
real rally or one-sided?
251 - **STRATEGIC (user, 2026-06-06): paid-Gemini as a cost-controlled QUALITY fallback.**
User OK slicing
252 videos <1GB + paid Gemini to build a stronger baseline NOW + as a product "calculated-
bit-expensive,
253 quality-significantly-improving" fallback. KEY: the hybrid Phase-3 ALREADY does this
(WASB candidates ->
254 Gemini verifies each PADDED CANDIDATE CLIP, not whole video -> cost bounded by
#candidates*cliplen, not
255 duration); it died only on FREE-tier quota. So paid key re-enables it. Recommended
stack: WASB propose ->
256 gate A (free, cuts junk -> fewer Gemini calls) -> Gemini verify survivors (paid,
precision+boundaries).
257 ? PRIVACY TRADE-OFF: sending video to Gemini breaks the "vendors/AI never see user
uploads" privacy lever
258 ([[monetization-plan]]) ? fine for OWNED/test/golden video; for end users make it an
explicit opt-in
259 premium tier. More annotated videos arrive ~next week (-> leave_one_video_out for
honest cross-video).
260
261 ## GEMINI REFINE (2026-06-06) ? WASB candidates -> Gemini verify/refine (in progress)
262 - `backend/eval/gemini_refine.py` (NEW, uncommitted on master working tree; + provider
retry edits in
263 `backend/providers/gemini.py`). Reuses GeminiProvider + badminton prompt +

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extract_video_chunk +
264 make_predict_fn + write_segments_csv (no reinvent). Drives Gemini over SHORT WASB
candidate clips
265 (~baseline 45, padded 3s, reencoded ? tiny uploads ~450MB total, NOT GB chunks).
Gemini returns the
266 true rally per clip or [] (rejects between-point feeds it can see). **Envelope per
candidate** (1 WASB
267 candidate ? 1 rally) so Gemini fragments don't shatter a long rally below min_dur.
User has a PAID key
268 (env GEMINI_API_KEY only ? NEVER store the key in memory/disk/commits).
269 - **MODEL GOTCHA (verified 2026-06-06):** `gemini-2.5-flash` works but hits frequent
**503 "high demand"
270 spikes; `gemini-2.0-flash` is **404 (deprecated for generateContent)** despite
appearing in models.list.
271 ? use **`gemini-flash-latest`** (clean, no 503, current stable flash). Account also
has gemini-3.x /
272 gemini-3.5-flash (newer than my training). Added upload+generate retries (3x backoff)
to the provider;
273 per-call client per worker (genai client not thread-safe ? socket errors when shared).
274 - Smoke (6 cand) clean on flash-latest: tightens boundaries (e.g. WASB 21.3-27.5 ?
Gemini 23.8-25.8),
275 keeps long rally whole. Full 45-candidate run launched (task bkap9jri2) ->
output/adarsh_gemini_refined.csv.
276 - NEXT after run: compare Gemini-refined vs WASB baseline/baseline+gate (Gemini as
SILVER-GT to finally
277 quantify Adarsh); commit gemini_refine + provider retries as a PR. Recall ceiling =
WASB candidates
278 (Gemini only sees proposed windows; rallies WASB missed entirely won't appear).
279
280 ## PRECISION FINDING (2026-06-06): full-context Gemini > candidate-refine (envelope
over-merges)
281 - Added `full_video_rallies()` to gemini_refine + `--full-video` (whole clip, downscaled
1280, chunked
282 <=120s to stay under 500MB) + clip downscaling in extract_video_chunk (scale_width).
Reuses provider/prompt.
283 - **TestLargeVideo (5.3K, 100s) head-to-head:** candidate-refine = 5 rallies/49s;
**full-context = 7 rallies/
284 34.5s**. Refine collapsed the 0:23-0:43 region into ONE 22s blob; full-context split
it into 3 real rallies
285 (23.5-26.5, 31.5-33.5, 39-42.5) with dead time between. Root cause: WASB merge_gap=2.0
bundled 3 rallies
286 into 1 candidate, then my **envelope-per-candidate** collapsed Gemini's fragments back
into 1 ? swept ~13s
287 dead time into a "rally" = PRECISION LOSS. Full-context is a strict improvement (same
rallies, tighter, +
288 correct split). Files: output/testlarge_gemini_rallies.csv (refine),
output/testlarge_gemini_fullvideo.csv (full).
289 - ? **USER CONFIRMED (2026-06-06): full-context is correct** ? 0:23-0:43 IS three
rallies (24-26, 31-34,
290 39-42), not one blob; later rallies also right. Full-context is the precision-improved
default.
291 - ? **FIXED + SHIPPED ? PR #29** (`feat/gemini-refine`): dropped the over-merging
envelope; tiny
292 invisibility splits healed by gap-tolerant merge (<=1s), real between-rally gaps kept
separate. Provider
293 hardening (upload+503 retries, MAX_UPLOAD_MB=500 guard), extract_video_chunk
scale_width downscaling,
294 default model gemini-flash-latest, per-worker client. 7 tests, suite 192 green. Modes:
refine (WASB
295 candidates) + `--full-video` (whole clip, chunked<=120s, downscaled).
296 - ? Adarsh full-context RUNNING (task b9fs6aylf) -> output/adarsh_gemini_fullvideo.csv
(~7 chunks).
297 - Ensemble voting (N Gemini judgments/rally, majority+median boundaries) = next
precision lever if needed.
298 - ? Gemini sees user video = breaks privacy lever ([[monetization-plan]]); OK for

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owned/test video, opt-in
299 premium tier for end users.
300
301 ## ? TEACHER?STUDENT DISTILLATION (user's framing, 2026-06-06): Gemini=offline teacher,
LOCAL-ONLY WASB+gate=student
302 - GOAL: use Gemini (cloud, offline, one-time) as silver-GT to CALIBRATE the local-only
knobs so PRODUCTION
303 runs 100% local (no Gemini at runtime ? privacy lever intact, ~0 marginal cost).
Gemini quality distilled
304 into thresholds.
305 - BUILT: `backend/eval/calibrate_local.py` (+test, 2 pass). predict_fn =
trajectory_to_action_windows
306 (velocity/merge/min_dur) + rally_gate(min_crossings); grid 5x3x3x4=180; scores vs
Gemini silver labels;
307 `leave_one_video_out` across videos = honest cross-video precision.
BASELINE_LOCAL=(0.02,2.0,2.0,0).
308 **DEPENDS ON PR #28 (rally_gate)** ? built on branch `feat/local-calibration` (stacked
on feat/rally-start-gate).
309 - INPUTS: WASB trajectories (testlarge_traj.csv, adarsh_traj.csv on disk) + Gemini full-
context label CSVs
310 (output/testlarge_gemini_fullvideo.csv done=7 rallies;
output/adarsh_gemini_fullvideo.csv RUNNING b9fs6aylf).
311 - NEXT: when Adarsh labels land ? manifest of 2 videos ? run calibrate_local ? tuned
local cfg + honest LOVO
312 precision (baseline-untuned vs calibrated held-out). Commit/PR (note #28 dependency).
More videos next week
313 ? stronger LOVO.
314 - BRANCH STACK now: #27 code-map, #28 gate, #29 gemini-refine, feat/local-calibration
(on #28). Suggest user
315 merges #28+#29 so local-calibration rebases onto clean master.
316
317 ## ? RAN local calibration vs Gemini silver-GT (2 videos: TestLargeVideo=7 rallies,
Adarsh=36) ? HONEST result
318 - **Held-out (LOVO) ? the trustworthy number:**
319 . F1-optimized: baseline 0.438 ? calibrated **0.483** ±0.17, overfit gap +0.035 (small
?) = modest real lift.
320 . Precision-optimized: baseline 0.444 ? calibrated **0.320** ? ±0.10, gap **+0.245
(severe overfit)** = does NOT transfer.
321 - Per-video fit-on-self (optimistic): Adarsh P 0.489?0.731 / F1 0.543?0.683;
TestLargeVideo barely moves (F1 0.33).
322 - **CONCLUSION (honest, don't oversell):** local-only pipeline agrees with Gemini only
~0.44 P/F1; threshold-
323 tuning hits a ceiling fast and OVERFITS for precision with only 2 dissimilar videos
(5.3K phone vs Adarsh);
324 they want opposite configs. TestLargeVideo also RECALL-capped (WASB ~5 windows vs 7
Gemini rallies).
325 - **REAL LEVERS for local precision (next):** (1) MORE teacher-labeled videos (next
week) ? stable LOVO;
326 (2) LEARNED distilled model ? train existing `classifier.py` on Gemini labels using
candidate features
327 (peak/mean velocity, active_frame_count, net-crossings) instead of 4 thresholds; (3)
raise WASB recall.
328 Recommend learned-distillation over more threshold-tuning.
329
330 ## CODE_MAP artifact ? PR #27 (`docs/code-map`, OPEN) ? see [[code-map-artifact]]
331 `docs/CODE_MAP.md` = dense cold-ramp navigation map
(pipeline/subsystems/contracts/gotchas/ramp-paths/
332 test-map). Built by a map->synthesize->adversarial-verify Workflow; verify caught real
errors (validator
333 order, nonexistent config.json indexing.wasb, extract_yolo_features gotcha). Regenerate
via the
334 `build-code-map` workflow. Read it FIRST for code ramp-up.
335
336 ## ? NEXT-SESSION pickup
337 - **User to eyeball** the tuned segments (`output/wasb_tuned_segments_vs_golden.csv`;

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regenerate via
338 `python -m backend.eval.export_wasb_segments --trajectory
C:\?\Temp\sample_long_traj.csv --labels
339 "Annotation Setup\golden_labels_badminton.csv" --fps 59.94 --frame-width 1920
--baseline`; add
340 `--video <mp4> --slice` for clips). Verdict drives whether to tune merge_gap
(fragmentation) /
341 min_window_duration (short rallies).
342 - **Real precision verdict** still needs a **2nd labelled video** ?
`leave_one_video_out`.
343 - Optional housekeeping: stale MERGED remote branches `feat/rally-slicer` (#20) +
`feat/wasb-substrate`
344 (#21) still on origin (never deleted; harmless, all on master). NOT deleted this
session given the
345 #21-branch-delete-lost-calibrate_wasb history ? ask before pruning. KEEP
346 `feat/khelacharya-shot-intelligence` (PR #11 closed, intentionally retained).
347
348 ## ? (1) EXPORT TUNED SEGMENTS DONE ? PR #26 MERGED to master (`feat/wasb-export-
segments`)
349 `backend/eval/export_wasb_segments.py` (+5 tests, suite 187 green). Reuses
make_predict_fn +
350 metrics.match_intervals/score + rally_slicer (exported CSV is in the golden/slicer
schema ? feeds
351 straight into clip cutting via `--slice --video`). Ran on REAL trajectory + golden
labels:
352 - **34 tuned segments** (cfg 0.05/1.0/1.0); vs 25 golden: TP 22, FN 3, FP 12 ? precision
0.647,
353 recall 0.880, **F1 0.746**, meanIoU 0.753 (reproduces calibration EXACTLY). Mean
boundary err:
354 start 0.64s / end 0.91s. Artifacts in `output/` (gitignored):
`wasb_tuned_segments.csv`,
355 `_baseline.csv`, `_vs_golden.csv`.
356 - **EYEBALL FINDINGS (honest):** recall strong, boundaries mostly <1s, but precision
0.647 = tuned
357 config OVER-segments. The 3 "misses" + 12 "FPs" are partly artifacts not pure errors:
358 · golden 3 (44.5-51.1s) FRAGMENTED into 2 sub-segments (44.8-46.9 + 48.4-50.8) ?
neither hits
359 IoU 0.5 ? 1 miss + 2 FPs from ONE rally (a merge_gap problem).
360 · golden 8 (118.1-119.6, 1.5s) detected as 118.1-123.0 but boundary too long ? 1 miss
+ 1 FP.
361 · golden 7 (95.6-98.0, 2.4s) genuinely dropped (short rally).
362 · boundary overruns: golden 1 start +6.16s (missed first 6s of a long rally), g21 end
+4.76s,
363 g23 end +3.05s.
364 · ~9 genuine extra segments to eyeball: 3.35-7.96, 36.9-39.4, 53.9-58.4, 60.5-62.9,
79.5-81.2,
365 107.2-111.2, 144.5-146.6, 188.2-190.3, 339.1-343.3 (warmup / between-point motion /
real
366 unlabeled rallies?).
367 · LEVERS if user wants better precision: raise merge_gap (fix fragmentation), revisit
368 min_window_duration for short rallies; real precision verdict needs a 2nd labelled
video
369 (`leave_one_video_out`).
370
371 ## ? PR status: #25 (checkpointing) + #26 (export) BOTH MERGED to master. No open
indexer PRs.
372
373 ## NEW session 2026-06-06 (late) ? user asked for tasks in order: (4) repo cleanup ? (2)
WASB checkpointing ? (1) export tuned segments
374 - ? ** (4) repo cleanup DONE:** collector local checkout was stranded on the merged
`feat/validate-delivery`
375 branch ? synced to `master` (ff to d68c5da = PR #9 merge) + deleted the merged branch.
Both repos now
376 genuinely clean on master. Untracked local `data/roora_*` artifacts left in place (not
committed).

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377 - ? **PR #11 (KhelAcharya shot-intel) CLOSED, branch retained.** Correction: there is NO
separate
378 khelacharya GitHub repo ? that folder is a 2nd clone of THIS repo. Separation is
branch-level; #11
379 kept shot-work off master. Closed to declutter; `feat/khelacharya-shot-intelligence`
branch persists,
380 reopenable. See [[sibling-internship-repo]] (corrected).
381 - ? ** (2) WASB-pass checkpointing DONE ? PR #25 MERGED to master** (`feat/wasb-
checkpointing`).
382 `wasb_infer.py` refactored: per-window detector cache `<out>_wasbcache/det_raw.jsonl`
(fsyc every
383 --flush-every=1000 windows) + atomic `manifest.json`; resume replays cache + skips
DataLoader to first
384 un-cached window; tracker re-run over full ordered cache (cheap, deterministic);
atomic tmp+replace
385 writes; trailing-corruption truncation; auto-invalidate on input change; idempotent
extract_frames.
386 11 new unit tests (`tests/test_wasb_cache.py`), suite **182 green**. WSL-validated:
tracker-only re-run
387 byte-identical; incremental resume matches full run except 3/300 rows ?6e-5 px (cuDNN
float noise from
388 re-batching, NOT a bug); --limit change correctly invalidates. Runner gets resume for
free (cache
389 derives from --out, not %TEMP%). **GOTCHA:** must re-stage `wasb_infer.py` to
`~/models/WASB-SBDT/src/`
390 in WSL when running directly (runner does this automatically).
391 - ? **NEXT: (1) export tuned segments** for user eyeball ("are rallies better?"). Use
tuned cfg
392 (velocity 0.05, merge_gap 1.0, min_dur 1.0) on the cached trajectory ? cut rally
clips.
393 Artifacts confirmed intact: `~/clips/sample_long_traj.csv` (21,284 frames) + Windows
Temp copy;
394 21,284 frames in `Temp/sample_long_frames`; golden labels at `Annotation
Setup\golden_labels_badminton.csv`.
395
396 ## Merged to master (both repos clean, branches deleted)
397 - collector #6 (CVAT?canonical adapter), #7 (forced/unforced vocab), indexer #19 (docs).
398 - Local master pulled + clean in both repos.
399
400 ## WASB inference API (reverse-engineered 2026-06-06 ? for the wrapper)
401 Repo `~/models/WASB-SBDT` (Hydra configs). Usage: `cd src && python3 main.py --config-
name=eval
402 dataset=badminton model=wasb
detector.model_path=./pretrained_weights/wasb_badminton_best.pth.tar`.
403 Inference core (`src/runners/eval.py::inference_video`): build detector + tracker +
dataloader ?
404 loop batches `(imgs, hms, trans, ...)` ? `detector.run_tensor(imgs, trans)` ? per-frame
preds ?
405 `tracker.refresh()` then `tracker.update(preds)` per frame ? `result_dict[img_path] =
{x,y,visi,score}`.
406 That dict IS the trajectory we want.
407 - `TracknetV2Detector(cfg)` (`src/detectors/detector.py`): builds model via
`build_model(cfg)`,
408 loads `cfg.detector.model_path` (`checkpoint['model_state_dict']`), DataParallel,
eval. Needs
409 CUDA (asserts GPU). `frames_in`=cfg.model.frames_in, `input_wh`=(model.inp_width,
inp_height).
410 `run_tensor(imgs, affine_mats)` ? model ? postprocessor ? per-frame (x,y,visi,score).
411 Preprocessing helpers: `dataloaders.build_img_transforms/read_image/get_transform`,
412 `utils.image.get_affine_transform/affine_transform`.
413 - eval.yaml defaults: model=wasb, detector=tracknetv2, tracker=online,
transform=default.
414 - **Wrapper plan:** extract video frames ? build `frames_in` sliding windows + affine
`trans`
415 (mirror dataset `__getitem__`) ? run detector+tracker ? emit CSV

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`Frame,Visibility,X,Y`
416 (EXACTLY what existing `tracknet_runner.parse_trajectory_csv` +
`trajectory_to_action_windows`
417 already consume ? big reuse). Use weights `wasb_badminton_best.pth.tar` (NOT
monotrack=noncommercial).
418 - **Build steps:** (1) `backend/detectors/wasb_infer.py` ? **DONE + VALIDATED** (branch
419 `feat/wasb-substrate`, pushed, not yet PR'd). Runs in WSL `wasb` env; reuses WASB
ImageDataset
420 transform + detector + online tracker; emits Frame,Visibility,X,Y. Verified on
rally-18 clip
421 (121 frames ? 107 visible, X439-1036/Y64-586, smooth = real shuttle track).
422 **GOTCHAS:** must run from `~/models/WASB-SBDT/src` in `wasb` env; Hydra overrides
need
423 `runner.gpus=[0]` (single-GPU box) + `runner.device=cuda`; `__getitem__` lives in
424 `dataloaders/dataset_loader.py::ImageDataset` (samples =
`{'frames':[paths],'annos':[{frame_path,
425 center:Center(is_visible,x,y)}}]`); read frames via /mnt (Google-Drive G: extraction
done on
426 Windows ffmpeg). To run: `conda activate wasb && cd ~/models/WASB-SBDT/src && python
wasb_infer.py
427 --frames_dir <dir> --weights ../pretrained_weights/wasb_badminton_best.pth.tar --out
<csv>`.
428 (2) `wasb_runner.py` ? DONE (Win adapter: stage video+script to WSL ? run `wasb` env ?
CSV;
429 REUSES `TrackNetRunner.parse_trajectory_csv`/`trajectory_to_action_windows` as
staticmethods ?
430 those are class staticmethods, NOT module funcs; only
`to_wsl_mnt_path`/`TrajectoryPoint` are
431 module-level). (3) `wasb_hybrid.py` ? DONE (mirrors tracknet_hybrid; registered
"wasb_hybrid";
432 reads thresholds from idx_cfg.wasb). All on branch `feat/wasb-substrate` (pushed, NOT
PR'd).
433 Suite 159 green; registry lists wasb_hybrid. **NOT yet end-to-end run**: full-video
WSL pass +
434 AI handoff untested (wasb_infer core IS validated on real frames). config.json needs
an
435 `indexing.wasb` section (defaults exist in WasbConfig). (4) per-video calibration
Workflow
436 = DEFERRED (user will call it). Calibration only needs Phase-2 windows (no AI
handoff/API).
437 Preprocessing (RESOLVED): `configs/model/wasb.yaml` ? name=hrnet, frames_in=3,
frames_out=3,
438 inp 512x288 (WxH). `dataloaders/__init__.py build_img_transforms`: T.ToTensor +
Normalize
439 mean[0.485,0.456,0.406]/std[0.229,0.224,0.225] (ImageNet). Resize via affine
(`utils.image.
440 get_affine_transform` ? input_wh; the affine matrix is the `trans`/`affine_mats`
passed to
441 run_tensor and used to map heatmap coords back to original frame). So wrapper: per
sliding window
442 of 3 consecutive frames ? affine-resize to 512x288 ? ToTensor+Normalize ? stack ?
detector.run_tensor
443 ? tracker ? (x,y,visi,score) per frame ? CSV Frame,Visibility,X,Y. Weight variant:
wasb_badminton_best.
444
445 ## GOAL CORRECTION (user clarified 2026-06-06 ? important)
446 User wants to "make the WASB pretrained WEIGHTS more robust with the 1 annotated video."
Honest reality:
447 - Our annotation = **temporal rally [start,end]** labels, NOT per-frame shuttle (x,y).
WASB is a shuttle
448 DETECTOR (needs x,y heatmap supervision). **So the rally labels CANNOT fine-tune
WASB's detector weights.**
449 - What the 1 video CAN quantifiably improve = the **rally-SEGMENTATION layer on top of
WASB**: the 3
450 windowing thresholds (velocity_thresh/merge_gap/min_window_duration) + WASB variant

```

choice. Measure

451 IoU/F1 on HELD-OUT golden rallies (calibrate on K, test on rest). That's the "quantifiable improvement".

452 - To genuinely fine-tune WASB DETECTOR weights ? need per-frame shuttle (x,y) labels (a different

453 annotation, or pseudo-labels) ? separate track, not this annotation.

454 - Also: neural weights are found by backprop, not by agents "trying weights + coordinating". Agents are

455 the right tool for the CONFIG search (variant  $\times$  3 thresholds  $\times$  strategy), not weight space.

456

457 ## Tuning plan = a Workflow (per user's ultracode ask), runs AFTER the wrapper exists

458 Once WASB trajectories exist: run a parallel **\*\*calibration Workflow\*\*** ? agents independently search

459 {WASB variant  $\times$  velocity\_thresh  $\times$  merge\_gap  $\times$  min\_window\_duration  $\times$  search-strategy}, each scored on

460 held-out golden rally IoU, coordinate ? pick max-precision config. (Can't run before a trajectory exists.)

461

462 ## ? PLAN: checkpoint the heavyweight WASB pass (resume across restarts) ? IMPLEMENT NEXT SESSION

463 Problem: the GPU detector pass (21k forward passes) writes the trajectory ONLY at the end, so a

464 reboot mid-pass loses all of it (happened 2026-06-06).

465 Key insight: **\*\*decouple the expensive GPU detector pass (checkpoint incrementally) from the cheap,**

466 **stateful CPU tracker pass (just re-run from the cache).\*\*** The online tracker can't resume

467 mid-stream, but if per-frame DETECTOR outputs are cached, the tracker re-runs over them in seconds.

468 Plan (prioritised):

469 1. **\*\*Stable cache dir keyed by video MD5\*\*** (not %TEMP%): `output/wasb\_cache/<md5>/` with `frames/`,

470 `det\_raw.jsonl` (incremental detector outputs), `trajectory.csv`, `manifest.json`

471 {video\_md5, frames\_total, frames\_done, cfg}. (Ties to the indexer MD5-keying gotcha.)

472 2. **\*\*Idempotent extraction:\*\*** skip if frame count already == expected.

473 3. **\*\*Incremental detector cache (the big win):\*\*** in wasb\_infer's batch loop, append each batch's

474 per-frame raw detections to `det\_raw.jsonl` and FLUSH every ~1000 frames; on startup read

475 manifest/det\_raw ? skip already-done frames ? resume. Loss bounded to <1000 frames (~secs of GPU).

476 4. **\*\*Tracker re-run:\*\*** once all frames are detected (fresh or resumed), run the tracker over the full

477 `det\_raw.jsonl` in order ? `trajectory.csv`. Cheap, deterministic, re-runnable.

478 5. **\*\*Atomic writes\*\*** (write .tmp then rename) so a crash mid-write can't corrupt the cache.

479 6. (Optional) expose as one resumable CLI stage.

480 Effort: moderate ? mostly a wasb\_infer refactor (incremental det cache + resume); the tracker

481 decoupling is the core design. Calibration grid stays cheap (operates on the cached trajectory) ?

482 no checkpointing needed there.

483

484 ## ? STEP 4 ? FIRST CALIBRATION DONE (2026-06-06) ? RESULT

485 Full WASB trajectory of the long badminton video: 21,284 frames, 37% visible-shuttle (sane).

486 `calibrate\_wasb` on 25 golden rallies (IoU 0.5):

487 - **\*\*baseline\*\*** thresholds (0.02,2.0,2.0): precision 0.577, recall 0.600, **\*\*F1 0.588\*\***, meanIoU 0.765.

488 - **\*\*tuned\*\*** (0.05,1.0,1.0): **\*\*F1 0.746\*\*** ? precision +12.2%, **\*\*F1 +26.8%\*\*** (OPTIMISTIC ? grid fit on full GT).

489 - **\*\*HONEST** (5-fold cross-val): held-out recall  $0.88 \pm 0.16$ , generalization\_gap +0.000\*\* (no overfit).

490 - Caveats: +X% is optimistic upper bound; honest within-video metric is RECALL;

```

precision/F1
491 generalization needs a 2nd labelled video ? `leave_one_video_out`. std ± 0.16 = small
sample.
492 - Driver `backend/eval/calibrate_wasb.py` was LOST from master in #21's branch-delete ?
RECOVERED
493 in PR #24 (merged). Trajectory cached at `~/clips/sample_long_traj.csv` (+ `C:\?\Temp\
494 sample_long_traj.csv`); frames at `C:\?\Temp\sample_long_frames` (4.1GB, deletable).
495 - NEXT: export tuned segments for user manual-verify; 2nd labelled video for honest
precision;
496 implement WASB-pass checkpointing (plan below); optional multi-agent tuning Workflow.
497
498 ## (historical) STEP 4 IN PROGRESS (2026-06-06)
499 - PR ##21 open (feat/wasb-substrate: substrate + calibration.py + `calibrate_wasb.py`
driver +
500 HOW_RALLY_DETECTION_WORKS doc). Suite 171 green.
501 - **Calibration driver DONE + tested:** `backend/eval/calibrate_wasb.py` (grid over
502 velocity_thresh/merge_gap/min_window_duration ? improvement_report + cross_validate).
Run:
503 `python -m backend.eval.calibrate_wasb --trajectory ~/clips/sample_long_traj.csv (copy
to Win)
504 --labels "<Annotation Setup>/golden_labels_badminton.csv" --fps 59.94 --frame-width
1920`.
505 - **GOTCHA:** WSL CANNOT see Google Drive `/mnt/g` (first pass b04odl4jl failed: "cannot
stat
506 /mnt/g/..."). Fix: copy the video G:C: on the Windows side first (`cp "/g/My
Drive/.../Sample
507 Long Video - Baadminton.MP4" /c/Users/avidu/AppData/Local/Temp/sample_long.mp4`, ~9s,
cached),
508 then WSL reads it via `/mnt/c`.
509 - **PERF GOTCHA:** wasb_infer's cv2 `imwrite` PNG extraction is pathologically slow
(~5fps ? ~1hr
510 for 21k frames, GPU idle the whole time). FIX: extract with **ffmpeg** (Windows, reads
the C:
511 copy): `ffmpeg -i C:\...\sample_long.mp4 -q:v 2 C:\...\sample_long_frames\%08d.jpg`
(21284 frames
512 in **3.5 min**, 4.1GB JPG). Then `wasb_infer --frames_dir` (now defaults
**batch_size=8,
513 num_workers=4** ? GPU ~80-90%). ffmpeg NOT in WSL; use Windows ffmpeg. wasb_infer
globs *.jpg too.
514 - **RESTART RECOVERY (2026-06-06):** laptop rebooted mid-inference. The 21,284 extracted
frames
515 (`C:\Users\avidu\AppData\Local\Temp\sample_long_frames`) + staged video survived (Temp
persisted);
516 trajectory CSV was NOT written (writes only at end). Resume = just RE-RUN the
inference command
517 below on the existing frames (NO re-extraction). Relunched as task brmhzav6x. The
518 `wasb_infer.py` batch/workers change is uncommitted on master working tree (commit
after verify).
519 - **Full-video WASB inference RUNNING** (task brmhzav6x; orig b4pqhqkte died on reboot):
`wasb_infer.py --frames_dir
520 /mnt/c/Users/avidu/AppData/Local/Temp/sample_long_frames --weights
521 ../pretrained_weights/wasb_badminton_best.pth.tar --out ~/clips/sample_long_traj.csv
522 --batch-size 8 --num-workers 4`. GPU confirmed ~80-90%, 6.4/8GB. Writes CSV at the
end. When done:
523 copy CSV to Windows (or read /mnt) ? run the driver above ? the "+X%" + cross-val.
Then clean
524 up `~/clips/sample_long_frames` (~30-40GB). Calibrate ONLY on the labelled Sample Long
Video
525 (only one with golden labels); a 2nd labelled video later ? leave_one_video_out.
526
527 ## NEXT-SESSION RUNBOOK ? step 4 (the actual calibration run)
528 Substrate (wasb_infer/runner/hybrid) + eval harness (calibration.py) are DONE on
529 `feat/wasb-substrate` (pushed). To submit:
530 1. **Full-video WASB trajectory (the slow long pole ? do first, cache it).** Untested at
full

```



```

531 scale; expect a debug lap. Two paths:
532 (a) via runner: `WasbRunner(WasbConfig()).run_predict(<video>, "output/wasb")` ?
stages 2.8GB
533 into WSL (off G: = possibly slow/flaky) + extracts frames + runs. OR
534 (b) safer: extract frames on Windows (ffmpeg from G:, proven) ? run in WSL:
535 `conda activate wasb && cd ~/models/WASB-SBDT/src && python wasb_infer.py
--frames_dir <dir>
536 --weights ../pretrained_weights/wasb_badminton_best.pth.tar --out
~/clips/full_traj.csv`.
537 Watch: disk for ~21k PNGs, GPU mem, /mnt read speed. Validate visible-shuttle % is
high during
538 known rallies, ~0 during dead time.
539 2. **GT** from `Annotation Setup/golden_labels_badminton.csv` ? rally [start,end] in
seconds
540 (use the verified rows; skip the 1 missed row whose end is still TBD = rally 7.5).
541 3. **Write the calibration driver** (not yet written): `predict_fn(cfg)` =
parse_trajectory_csv(full_traj)
542 ? `TrackNetRunner.trajectory_to_action_windows(points, fps=59.94, frame_width=1920,
**cfg)` ?
543 [start,end]; cfg = {velocity_thresh, merge_gap, min_window_duration}. Grid over
those.
544 4. **Run** `backend/eval/calibration.py`: `improvement_report(predict_fn, baseline_cfg,
best_cfg, GT)`
545 for the "+X%"; `cross_validate` (within-video recall + gap) for the overfit check.
(Only 1 video
546 now ? leave_one_video_out waits for more videos = the real generalization metric.)
547 5. Optional: the **multi-agent tuning Workflow** (token-heavy) instead of a plain grid
in step 3.
548
549 ## Eval/overfit harness BUILT + license soul-check (2026-06-06)
550 - `backend/eval/calibration.py` (on `feat/wasb-substrate`, 9 tests; segmenter-agnostic
predict_fn):
551 `improvement_report` = baseline-vs-tuned %? precision/F1 (the "+X%"); `cross_validate`
=
552 within-video k-fold, default metric **recall** (global windows vs a GT *subset* can't
fairly
553 score precision), reports `generalization_gap` (train?test = overfit alarm);
`leave_one_video_out`
554 = the honest metric as videos arrive (pick config on others, score held-out). This is
the
555 overfitting guard for "more videos coming."
556 - **LICENSE:** CODE clean ? WASB-SBDT MIT (verified), torchvision/supervision
permissive, AGPL
557 Ultralytics gone (PR #8), MonoTrack never referenced (default = wasb_badminton_best).
?? OPEN:
558 WASB *weights* trained on academic NYCU/TrackNetV2 data ? confirm terms or retrain
before
559 COMMERCIAL deploy (fine for R&D). See [[tracknet-wasb-wsl2-decision]].
560
561 ## Step-3 substrate finding (2026-06-06)
562 - WASB IS the plan and is ~ready: `wasb` WSL env + WASB-SBDT repo + pretrained badminton
weights
563 all present (verified); healthcheck OK (TF2.17/GPU). The wired `tracknet_runner`
targets
564 TrackNetV4 = NOT runnable (no weights, repo not cloned). To use WASB: build inference
wrapper +
565 runner adapter + `wasb_hybrid` segmenter, then calibrate
`velocity_thresh/merge_gap/min_window_duration`
566 vs golden labels (eval held-out). See [[tracknet-wasb-wsl2-decision]] STATUS
2026-06-06.
567
568 ## Merged: #6, #7, #19, #8, #20, **#9 (validate-delivery)**
569 - collector master has: adapter, forced/unforced vocab, last_shot_outcome (two-field).
570 - indexer master has: docs mirror, rally slicer (`backend/tools/rally_slicer.py`).
571

```

```

572 ## Open PR (pushed, awaiting merge)
573 - **collector #9** `feat/validate-delivery` ? per-video QA gate. `curate validate-
delivery`
574 (CVAT or CSV) ? `_report.md`/`.json` + `_review.csv` (auto manual-verify
sheet);
575 exit!=0 on blocker. New `possible_missed_rally` gap check.
`docs/INGESTION_PIPELINE.md` FAQ.
576 104 tests; verified on real delivery (BLOCKED on 1?2 overlap, flags 2/7/16).
577
578 ## Pilot status ? DONE verifying
579 - User verified **all** rallies: only the previously-flagged rows are issues;
**everything else
580 is clean trainable data**. The [start,end] intervals are settled ? **segmentation GT
is ready**
581 (ending_reason/outcome specifics for flagged rows are metadata, not needed for
temporal seg).
582 - Golden master: `C:\Users\avidu\Projects\Annotation Setup\golden_labels_badminton.csv`
583 (24 + 2 missed = rallies; 7.5 @1:46.031 confirmed via screenshot; 16.5 = 3:34?3:43).
584
585 ## Decisions locked
586 - **Two-field ending**: `ending_reason` (why) + `last_shot_outcome` ? {in,net,out,n_a}
(where).
587 - **Accept Roora's CVAT export** ? adapter ingests it; do NOT ask them to switch to CSV.
588 - **Absorb timestamps** ? adapter recomputes `seconds = frame/fps`; mm:ss also parsed
everywhere
589 (indexer parse_timestamp / collector parse_time_string / compiler / slicer). Minor
slack
590 absorbed by VideoCompiler padding (0.5/1.0s). Don't ask Roora to fix time computation.
591 - Roora report (`Annotation Setup\Roora Pilot Review ? Badminton (2026-06-06).md`)
revised to
592 only ask for what code can't fix: **completeness (no missed rallies), one-box-per-
rally
593 boundaries, add `last_shot_outcome` dropdown, pickleball/TT scope.**
594
595 ## Next steps (sequence 1?2?3, user approved)
596 > User will get **5?6 30-min+ videos** annotated soon ? the **validation phase must be
ready**
597 > to auto-check each delivery against our conventions BEFORE that lands.
598 1. ? **last_shot_outcome vertical** ? MERGED (#8).
599 2. ? **validate-delivery framework** ? DONE (collector PR #9, awaiting merge). Label-
only gap
600 heuristic for missed rallies (a motion-based 2nd opinion over raw video is a future
enhancement).
601 3. (claude) **NEXT: per-video calibration loop** ? slicer merged (#20); tune segmenter
thresholds
602 on K labeled rallies, measure IoU on the HELD-OUT rest (no leakage). Then learned
fine-tuning.
603 4. (claude) Build segmentation GT from golden labels ? convert-cvat/import-local/export
(the
604 actual "train on my own video" step; needs the corrected golden labels, overlap
already resolved).
605
606 ## Open design threads
607 - Per-video few-shot adaptation: calibration first (static camera = good case), honest
eval
608 (held-out rallies). cf. train/eval guardrail in [[collector-indexer-bridge]].
609

```

## 65. user (2026-06-07T08:11:37.202Z)

```

1 ---
2 name: session-handoff
3 description: "Living \"you are here\" handoff for this project ? current state, next
steps, open decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key
decision/ADR, phase shift, session wrap); \"update the handoff\" forces an immediate full

```

```

rewrite."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: eb4b7b53-2f77-4c4d-83e1-4fa5e77f3e3f
8 ---
9
10 # Session handoff ? badminton-highlight-indexer (+ siblings: sports-data-collector,
rally-annotator)
11
12 **Last updated:** 2026-06-07 . **Refresh policy: AUTO-refresh at meaningful
checkpoints** ? PR merged,
13 key decision/ADR, phase shift, session wrap-up (NOT on-request-only). "update the
handoff" forces an
14 immediate full rewrite. **Staleness tripwire:** if ?3 substantive [[current-state]]
checkpoints ? or any
15 PR-merge / new ADR / phase shift ? have landed since the "Last updated" date above,
refresh this handoff now.
16 Pointer, not a duplicate. **For the freshest blow-by-blow state, read [[current-state]]
first**
17 (updated every micro-checkpoint); this handoff is the slower ~1-page orientation.
18
19 ## You are here (2026-06-07)
20 - **Rally-detection stack matured & all merged.** WASB shuttle-trajectory substrate is
**resumable/
21 checkpointed** (md5-ish cache, incremental detector flush, tracker re-run). On top: a
rally layer =
22 windowing thresholds + a **net-crossing "gate A"** + a **distilled classifier**
(`distill_local`).
23 Eval harness: `calibrate_wasb` / `calibration.py` (LOVO) / `metrics.py`.
24 - **Honest core finding:** WASB-only threshold tuning **ceilings ~F1 0.44** and the
learned distilled
25 model **underperforms the baseline at n=2 videos**. The blocker is **golden data +
WASB candidate
26 recall**, not the method. (Gemini-as-teacher gave us the first real measurement.)
27 - **Strategy decided (ADR-007):** Gemini = **offline teacher only** ; **AUDIO** hit-
detection is the next
28 local cue, **fused in our rally layer** (WASB stays a frozen black box) ? Option A.
**Pose/stance =
29 parked #2. **Option B = the eventual end-state**: one OWNED end-to-end multimodal
model
30 (frames+audio+pose?rallies) fine-tuned on our data, dropping WASB-weight dependence +
the recall ceiling.
31 - **Golden data path:** owner **self-rates** clips with the new **standalone VLC
annotator repo**
32 (github.com/avidullu/rally-annotator, MIT, multi-sport) ? CSV ? in-repo `--labels`
loaders / collector
33 `import-local`. Human labels > the Gemini silver labels the model was starving on.
34 - **? SESSION SUNSET (2026-06-07).** Indexer PRs **25?#35 merged**, master green (**209
tests**), tree clean
35 (only `feat/khelacharya-shot-intelligence` retained; **no open PRs**). Annotator repo
public/clean. **Audio E1
36 ran ? classical onset = ~2.2x discrimination ceiling, NOT adopted for fusion**
(detector kept; quantitative
37 gains/losses + reusable lessons in [[audio-e1-findings]]). Modeling **PARKED** pending
golden labels.
38 ? PR #36 (`docs/AUDIO_E1_FINDINGS.md`) MERGED to master (2026-06-07); record also in
[[audio-e1-findings]] + plan $10.
39
40 ## Next steps / open threads
41 1. **Owner:** rate a few clips in VLC (annotator v1.2 ? needs a live multi-sport smoke
test), and/or merge
42 nothing pending (none open).
43 2. **Audio E1 ? DONE ? PR #35. Verdict: classical-onset audio is too weak (~2.2x
ceiling).** Built

```

```

44 `backend/audio/hit_detector.py` + `audio_e1_probe.py` (numpy/ffmpeg, no scipy, 209
tests). On existing GoPro
45 audio vs Gemini silver: audio fires in 100% of rallies + recovers 100% of WASB-missed
rallies (real recall
46 signal), BUT discrimination only ~2.2x and audio-only F1 < WASB. **Band-pass (Option
A) tried ? does NOT help**
47 (HF bands worse; ceiling holds; NOT a warm-up artifact ? between-point
tosses/footwork/talk fool a classical
48 onset detector). **Next (user decides):** E2 learned TIMBRE classifier (needs labeled
hits ? only plausible way
49 past 2.2x) OR re-prioritize to the owner's "other hunch" / pose-stance #2 / more
golden data for WASB+distilled.
50 Details: AUDIO_RALLY_DETECTION_PLAN.md §10 + [[current-state]].
51 3. Stance heuristic (#2) after the owner's "other hunch"; Option B once the corpus is
healthy.
52 4. Cached artifacts LEFT IN PLACE for re-use (owner will rate):
`Temp\{adarsh,testlarge}_frames` + trajectories;
53 `Temp\verylarge_frames` (partial). Adarsh/TestLargeVideo trajectories are ready to
feed calibration.
54
55 ## Key decisions (don't relitigate without reason)
56 - **Audio now ? stance parked ? Option B eventual** (ADR-007). Modular cues are how we
earn Option B
57 (each is cheap/debuggable and generates the labels the end-to-end model needs).
58 - **Fusion is in OUR rally layer, never inside WASB** (WASB = frozen vision-only shuttle
detector; no audio
59 path; retraining needs x,y+audio labels we lack). Two modes: feature-level (?
distilled classifier) +
60 decision-level recall-recovery. (ADR-007 §Integration architecture.)
61 - **Local-only inference = privacy lever**; **permissive licensing only** (ADR-005);
Gemini paid+offline only.
62 - **Distilled model > threshold tuning** is the direction, but **needs more videos**
(n=2 overfits).
63 - Per-video adaptation = calibration on held-out; never mix train/eval. Frequent **small
PRs** + continuous
64 [[current-state]] updates (this handoff on request only).
65
66 ## Ramp-up kit (read these to get current)
67 - [[current-state]] (live blow-by-blow), this handoff, [[audio-e1-findings]] (audio
gains/losses + lessons), [[working-agreement-checkpoints]], [[working-agreement-attribution]],
[[feedback-tail-long-job-logs]].
68 - **CODE/architecture:** `docs/CODE_MAP.md` FIRST (dense map). See [[code-map-
artifact]].
69 - **Strategy/decisions:** `docs/DECISIONS.md` (esp. **ADR-007** audio),
`docs/RALLY_DETECTION_RESEARCH.md`
70 (cited methods survey), `docs/AUDIO_RALLY_DETECTION_PLAN.md`,
`docs/VIDEO_RECORDING_GUIDELINES.md` (audio now required).
71 - **Key code:** `backend/eval/{distill_local,calibrate_local,calibrate_wasb,gemini_refin
e,metrics,calibration}.py`,
72 `backend/detectors/{wasb_infer,wasb_runner,rally_gate,tracknet_runner}.py`.
73 - **Annotator repo:** `C:\Users\avidu\Projects\rally-annotator\` (+ installed at
`%APPDATA%\vlc\lua\extensions\`).
74 - Memory notes: golden-set-vendoring, collector-indexer-bridge, candidate-segments-
finetuning, sibling-internship-repo.
75

```

## 66. assistant (2026-06-07T08:12:50.749Z)

I now have the broader context. The memory shows this parser fix was a known **\*\*"DUP RISK"\*\*\*** (current-state.md lines 44?47) ? there were concerns about it being fixed twice. The merged PR #10 resolves that. Let me verify the PR numbers and confirm there's no double-merge before updating the memory accurately.

## 67. user (2026-06-07T08:12:56.311Z)

```

=== all PRs (collector) ===
[{"baseRefName":"master","headRefName":"feat/observability-
reports","mergedAt":null,"number":11,"state":"OPEN","title":"feat: per-video observability
reports (collector)"}, {"baseRefName":"master","headRefName":"fix/canonical-csv-zero-
start","mergedAt":"2026-06-07T08:08:51Z","number":10,"state":"MERGED","title":"fix(parser): keep
a rally that starts at 0.0s in
load_canonical_csv"}, {"baseRefName":"master","headRefName":"feat/validate-
delivery","mergedAt":"2026-06-06T10:19:27Z","number":9,"state":"MERGED","title":"feat(ingest):
validate-delivery ? per-video QA gate + structured report +
FAQ"}, {"baseRefName":"master","headRefName":"feat/last-shot-
outcome","mergedAt":"2026-06-06T10:07:06Z","number":8,"state":"MERGED","title":"feat(schema):
add last_shot_outcome (in/net/out/n_a) alongside
ending_reason"}, {"baseRefName":"master","headRefName":"refactor/ending-reason-forced-unforced",
mergedAt":"2026-06-06T08:48:59Z","number":7,"state":"MERGED","title":"refactor(annotation):
adopt forced_error/unforced_error ending_reason
vocabulary"}, {"baseRefName":"master","headRefName":"feat/cvat-rally-
adapter","mergedAt":"2026-06-06T08:48:45Z","number":6,"state":"MERGED","title":"feat(ingest):
CVAT rally export -> canonical CSV
adapter"}, {"baseRefName":"master","headRefName":"docs/annotation-
warmup","mergedAt":"2026-06-04T04:40:17Z","number":5,"state":"MERGED","title":"docs(annotation):
warm-up/knock-up exclusion + how-to-spot-it
tip"}, {"baseRefName":"master","headRefName":"docs/sync-after-
multisport","mergedAt":"2026-06-02T17:22:16Z","number":4,"state":"MERGED","title":"docs: sync
README + PIPELINE_ROLE after multi-sport +
cleanup"}, {"baseRefName":"master","headRefName":"chore/tech-debt-
cleanup","mergedAt":"2026-06-02T17:06:22Z","number":3,"state":"MERGED","title":"chore: tech-debt
cleanup (config loader, parsing utils, deps,
fixes)"}, {"baseRefName":"master","headRefName":"docs/annotation-templates","mergedAt":"2026-06-
02T16:38:26Z","number":2,"state":"MERGED","title":"docs(annotation): intake manifest + NDA
templates + TODO markers"}, {"baseRefName":"master","headRefName":"feat/pickleball-tabletennis-
ingestion","mergedAt":"2026-06-02T16:27:49Z","number":1,"state":"MERGED","title":"Add pickleball
& table-tennis rally ingestion + annotation docs + test coverage"}]
=== does master's history show the parser fix exactly once? (search merge commits) ===

99dfa22 Merge pull request #10 from avidullu/fix/canonical-csv-zero-start
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
be75d80 feat(ingest): CVAT rally export -> canonical CSV adapter
=== count commits touching the fix lines in master ===
f7cab11 fix(parser): keep a rally that starts at 0.0s in load_canonical_csv
2882a03 feat(ingest): validate-delivery ? per-video QA gate + structured report
810541e feat(schema): add last_shot_outcome (in/net/out/n_a) alongside ending_reason
aae93c0 fix(ingest): drop coarse_sampling issue from CVAT adapter
be75d80 feat(ingest): CVAT rally export -> canonical CSV adapter

```

## 68. assistant (2026-06-07T08:13:38.882Z)

Facts confirmed and clean:

- **PR #10** (``fix(parser)?``, branch ``fix/canonical-csv-zero-start``) ? **MERGED** to master (``99dfa22``).
- **PR #11** (``feat: per-video observability reports (collector)``) ? still **OPEN**, observability-only.
- master history shows the parser fix **exactly once** (``f7cab11``) ? **no double-merge**. PR #11 doesn't touch ``rally_cvat.py``, so it'll still merge cleanly.

This resolves the **"DUP RISK"** the memory was tracking. Let me update the checkpoint files. First, the two now-stale bullets in ``current-state.md``:

## 69. user (2026-06-07T08:13:44.002Z)

The file `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md` has been updated successfully. (file state is current in your context ? no need to Read it back)

## 70. user (2026-06-07T08:13:51.216Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

## 71. assistant (2026-06-07T08:14:24.919Z)

Now refresh `session-handoff.md` ? it's badly stale (still describes a pre-observability "session sunset" with "no open PRs / 209 tests"), while a major feature is mid-flight and a PR just merged. The staleness tripwire (PR-merge + many checkpoints since "Last updated") calls for a refresh. I'll base it strictly on the authoritative `current-state.md`.

## 72. assistant (2026-06-07T08:14:35.643Z)

---

```
name: session-handoff
description: "Living \"you are here\" handoff for this project ? current state, next steps, open decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision/ADR, phase shift, session wrap); \"update the handoff\" forces an immediate full rewrite."
metadata:
 node_type: memory
 type: project
 originSessionId: eb4b7b53-2f77-4c4d-83e1-4fa5e77f3e3f

```

# Session handoff ? badminton-highlight-indexer (+ siblings: sports-data-collector, rally

**\*\*Last updated:\*\*** 2026-06-07 (observability phase) · **\*\*Refresh policy:** AUTO-refresh at meaningful checkpoints ?  
PR merged, key decision/ADR, phase shift, session wrap-up (NOT on-request-only). "update the handoff" forces an immediate full rewrite. **\*\*Staleness tripwire:\*\*** if ?3 substantive [[current-state]] checkpoints ? or any PR-merge / new ADR / phase shift ? have landed since the "Last updated" date above, refresh this handoff now.  
Pointer, not a duplicate. **\*\*For the freshest blow-by-blow state, read [[current-state]] first\*\*** (updated every micro-checkpoint); this handoff is the slower ~1-page orientation.

## You are here (2026-06-07) ? ? OBSERVABILITY-REPORTS feature mid-flight (multi-session/concurrent)

- **\*\*Active feature:\*\*** always-on **\*\*per-video observability reports\*\*** across BOTH tools (indexer + collector),  
powered by a NEW shared lib **\*\*`sports-obsreport`\*\*** (pkg `obsreport`) ? a small "report config language" on permissive primitives (pydantic/Jinja2/PyYAML). Each tool emits a **\*\*technical\*\*** report (debuggable via README+FAQ)  
AND a **\*\*customer-friendly summary\*\*** (internals stripped). Reports written **\*\*next to the video\*\*** by default (override `report.output\_dir`). Approved plan: `C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md` (read it to resume).  
- **\*\*Status by piece:\*\***  
- ? **\*\*obsreport lib\*\*** built + on GitHub **\*\* (PRIVATE) \*\*** github.com/avidullu/sports-obsreport (main + tags v0.1.0?v0.1.3).  
Editable-installed into Python 3.13. ~74 lib tests green. Modules:  
envelope/spec/rules/render/io/recorder/diff/schema.  
- ? **\*\*Indexer Phase A\*\*** ? **\*\*PR #41 OPEN\*\*** (`feat/observability-reports`, rebased onto #39 typed-config `backend/config/`).  
`backend/reporting/` (soft-import, never-raises), wired CLI+API. My reporting: **\*\*226 tests pass.\*\***  
?? **\*\*2 PRE-EXISTING red tests from #39\*\*** (NOT mine, left untouched):  
`test\_config.py::test\_load\_default\_config`  
+ `test\_save\_default\_config\_roundtrip` (Path==str on Windows).  
- ? **\*\*Collector Phase B\*\*** ? **\*\*PR #11 OPEN\*\*** (`feat/observability-reports`, observability-

ONLY). `src/reporting/`  
 (adapter Issue?Flag + recorders), wired curate validate-delivery/convert-cvat/import-local.  
 \*\*112 tests pass.\*\*  
 - ? \*\*Collector parser zero-start fix\*\* ? \*\*PR #10 MERGED\*\* to master (`99dfa22`). Separate from the reporting PR;  
 DUP RISK resolved (fix landed once; PR #11 carries no parser fix). See [[current-state]].  
 - \*\*?? NEXT: Phase D ? debuggability eval\*\* (ultracode Workflow: fresh-reader agents over broken-run reports ? refine wording; render a report to an image + eyeball) + committed golden tests (faq\_ref-resolution + flag-set) in both repos.

## Standing project context (rally-detection stack ? stable, mostly merged; PARKED pending golden data)

- WASB shuttle-trajectory substrate is resumable/checkpointed; on top = rally layer (windowing thresholds +  
 net-crossing "gate A" + distilled classifier `distill\_local`); eval harness  
 `calibrate\_wasb`/`calibration.py` (LOVO).  
 - \*\*Honest core finding:\*\* WASB-only threshold tuning \*\*ceilings ~F1 0.44\*\*; learned distilled model \*\*underperforms  
 baseline at n=2 videos\*\*. Blocker = \*\*golden data + WASB candidate recall\*\*, not the method.  
 - \*\*Audio E1 ran ? classical-onset = ~2.2x discrimination ceiling, NOT adopted\*\* (band-pass didn't help; detector kept,  
 indexer PR #35 merged). Gains/losses + reusable lessons in [[audio-e1-findings]]. Next audio path = E2 learned timbre (needs labeled hits).  
 - \*\*Golden data path:\*\* owner self-rates clips in the standalone \*\*VLC annotator\*\*  
 (github.com/avidullu/rally-annotator,  
 MIT, multi-sport) ? CSV ? in-repo `--labels` loaders / collector `import-local`.

## Key decisions (don't relitigate without reason)

- \*\*Observability:\*\* always-on; reports next-to-video; technical + customer views from ONE typed envelope; shared  
 schema via the `obsreport` lib (reusable config language); build-backend MUST be  
 `setuptools.build\_meta` (underscore).  
 - \*\*Strategy (ADR-007):\*\* Gemini = offline teacher only; \*\*AUDIO\*\* was the next local cue,  
 \*\*fused in OUR rally layer\*\*  
 (WASB stays a frozen vision-only detector); \*\*pose/stance parked #2\*\*; \*\*Option B\*\* (one OWNED end-to-end multimodal  
 model) = eventual end-state ? modular cues are how we \*earn\* it (each generates the labels it needs).  
 - \*\*Local-only inference = privacy lever; permissive licensing only (ADR-005); distilled >  
 threshold tuning but needs  
 more videos (n=2 overfits); per-video adaptation = calibration on held-out, never mix train/eval.\*\*  
 - \*\*Working style:\*\* frequent \*\*small PRs\*\* + continuous [[current-state]] updates (this handoff auto-refreshed at checkpoints).

## Ramp-up kit (read these to get current)

- [[current-state]] (live blow-by-blow ? read FIRST), this handoff, [[working-agreement-checkpoints]],  
 [[working-agreement-attribution]], [[feedback-tail-long-job-logs]], [[audio-e1-findings]].  
 - \*\*Observability plan:\*\* `C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md`. Lib at  
 `C:\Users\avidu\Projects\sports-obsreport`.  
 - \*\*CODE/architecture:\*\* `docs/CODE\_MAP.md` FIRST (dense map). See [[code-map-artifact]].  
 - \*\*Strategy/decisions:\*\* `docs/DECISIONS.md` (esp. \*\*ADR-007\*\* audio),  
 `docs/RALLY\_DETECTION\_RESEARCH.md`,  
 `docs/AUDIO\_RALLY\_DETECTION\_PLAN.md`, `docs/VIDEO\_RECORDING\_GUIDELINES.md`.  
 - \*\*Key code:\*\* `backend/reporting/` (indexer) + `src/reporting/` (collector);  
 `backend/eval/{distill\_local,  
 calibrate\_local,calibrate\_wasb,gemini\_refine,metrics,calibration}.py`;  
 `backend/detectors/{wasb\_infer,wasb\_runner,rally\_gate,tracknet\_runner}.py`.  
 - Memory notes: golden-set-vendoring, collector-indexer-bridge, candidate-segments-finetuning,  
 sibling-internship-repo, monetization-plan.

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 74. assistant (2026-06-07T08:14:53.941Z)

Done. PR #10 is merged and everything's reconciled and recorded.

#### Post-merge wrap-up ? `sports-data-collector`

**\*\*Verified merged:\*\***

- **\*\*[#10 `fix(parser): keep a rally that starts at 0.0s in load\_canonical\_csv`](https://github.com/avidullu/sports-data-collector/pull/10)\*\*** ? MERGED to `master` (merge commit `99dfa22`, by avidullu).
- master history shows the parser fix **\*\*exactly once\*\*** (`f7cab11`) ? **\*\*no double-merge\*\***.

**\*\*Cleanup done:\*\***

- Local `master` synced to `99dfa22` (fast-forwarded without disturbing your working tree).
- Merged local branch `fix/canonical-csv-zero-start` deleted; remote branch was auto-deleted by GitHub on merge (the fetch also pruned several other already-merged branches).
- Your working tree is left on `feat/observability-reports` (only unrelated untracked `data/` artifacts remain).

**\*\*Resolved the "DUP RISK" the memory was tracking:\*\*** the zero-start bug was at risk of being fixed twice (a duplicate `fix(parser)` commit lived on the observability branch + this dedicated branch). The dedicated branch won via PR #10; the duplicate was dropped, so the still-open **\*\*[#11 per-video observability reports](https://github.com/avidullu/sports-data-collector/pull/11)\*\*** carries no parser fix and doesn't touch `rally\_cvat.py` ? it'll merge cleanly.

**\*\*Memory checkpoint updated\*\*** (PR-merge trigger, per your working agreement):

- `[current-state.md](C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md)` ? flipped the DUP RISK bullet to ? RESOLVED and recorded PR #10 merged.
- `[session-handoff.md](C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md)` ? refreshed (it was stale at a pre-observability "session sunset"); now reflects the in-flight observability-reports phase with PRs #41/#11 open and #10 merged.

I did **\*\*not\*\*** create a memory file for the fix itself ? it lives in git history, which your memory convention says not to duplicate.

The one open item in this repo is **\*\*PR #11 (observability)\*\***, which is the other session's work ? I've left it untouched.

#### 75. user (2026-06-07T08:14:57.230Z)

Is there a clean way of running all the tests in the repo?

Can you see if <https://github.com/avidullu/badminton-highlight-indexer/pull/41> conflicts can be resolved and resynced to the new directory structure?

#### 76. assistant (2026-06-07T08:15:16.919Z)

Two tasks here, both in the **\*\*indexer\*\*** repo (current working dir). Let me scout the current state before doing anything ? test setup, PR #41's conflict status, and what the "new directory structure" actually is.

#### 77. user (2026-06-07T08:15:24.182Z)

=== current branch + status ===

feat/observability-reports

=== PR #41 mergeable state ===

```
{
 "additions": 783,
 "baseRefName": "master",
 "changedFiles": 9,
 "deletions": 69,
 "headRefName": "feat/observability-reports",
 "mergeStateStatus": "DIRTY",
 "mergeable": "CONFLICTING",
 "number": 41,
 "state": "OPE
```



```
N", "title": "feat: per-video observability reports (indexer) + shared obsreport lib"}
=== local vs origin/master ===
From https://github.com/avidullu/badminton-highlight-indexer
- [deleted] (none) -> origin/docs/audio-e1-findings
 ce6a0e9..0355417 master -> origin/master
0355417 Merge pull request #40 from avidullu/refactor/approved-directory-structure
1e96e07 test: fix mock patch targets after structure refactor and config test assertions
8615534 refactor: adopt approved layered directory structure
=== merge-base of HEAD and origin/master ===
ce6a0e9188385b79a3e59a049b3a001f70fb288c
fa60efa test(reporting): cover typed AppConfig normalization in emit (post-#39 rebase)
02aa99a chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
f1839b8 feat(reporting): always-on per-video observability report (indexer)
```

#### 78. user (2026-06-07T08:15:24.319Z)

```
pytest.ini
requirements.txt
```

#### 79. user (2026-06-07T08:15:24.335Z)

No files found

#### 80. assistant (2026-06-07T08:15:57.745Z)

Important finding: PR #41 is **CONFLICTING**. The "new directory structure" is **PR #40** (``refactor/approved-directory-structure``), which just merged to master (``0355417``) ? it adopted a "layered directory structure" **after** the observability branch diverged (merge-base is the old master ``ce6a0e9``). So the branch needs resyncing onto the restructured tree.

Let me understand the test setup, the branch's commits, and exactly what the refactor moved.

#### 81. user (2026-06-07T08:15:58.701Z)

```
1 [pytest]
2 # When `pytest` is invoked without an explicit path, only collect from tests/.
3 # scratch/ holds debugging scripts that hit real APIs (Gemini upload, etc.)
4 # and should never run as part of the regular test suite.
5 testpaths = tests
6
```

#### 82. user (2026-06-07T08:16:03.088Z)

```
=== commits on branch beyond old-master merge-base (ce6a0e9..HEAD) ===
fa60efa test(reporting): cover typed AppConfig normalization in emit (post-#39 rebase)
02aa99a chore(reporting): per-tool customer_verdict + bump obsreport pin to v0.1.3
f1839b8 feat(reporting): always-on per-video observability report (indexer)
=== local HEAD vs origin/feat/observability-reports ===
fa60efaf0050abc431e01293edce103d9ee836ce
fa60efaf0050abc431e01293edce103d9ee836ce
=== files changed by the branch (vs merge-base) ===
 README.md | 17 ++++
 backend/main.py | 168 ++++++-----
 backend/reporting/__init__.py | 71 ++++++
 backend/reporting/harvest.py | 195 ++++++
 backend/reporting/spec.yaml | 117 ++++++
 backend/validators/base.py | 15 +++-
 requirements.txt | 7 ++
 tests/test_reporting_harvest.py | 190 ++++++
 tests/test_validators_base.py | 72 ++++++
 9 files changed, 783 insertions(+), 69 deletions(-)
=== what PR #40 restructure moved (rename detection) ===
 backend/{ => api}/static/app.js | 0
 backend/{ => api}/static/index.html | 0
```

|                                                       |                  |
|-------------------------------------------------------|------------------|
| backend/{ => api}/static/style.css                    | 0                |
| backend/eval/audio/e1_probe.py                        | 2 +-             |
| backend/eval/calibrate_local.py                       | 4 +-             |
| backend/eval/calibrate_wasb.py                        | 2 +-             |
| backend/eval/distill_local.py                         | 4 +-             |
| backend/eval/export_wasb_segments.py                  | 4 +-             |
| backend/eval/harness.py                               | 2 +-             |
| backend/eval/regression.py                            | 2 +-             |
| backend/indexer/__init__.py                           | 16 ----          |
| backend/{ => infrastructure}/database.py              | 0                |
| backend/main.py                                       | 18 +----         |
| backend/{ => pipeline}/audio/__init__.py              | 0                |
| backend/{ => pipeline}/audio/hit_detector.py          | 0                |
| backend/{ => pipeline}/detectors/__init__.py          | 0                |
| backend/{ => pipeline}/detectors/rally_gate.py        | 0                |
| .../{ => pipeline}/detectors/tracknet_runner.py       | 0                |
| backend/{ => pipeline}/detectors/wasb_infer.py        | 0                |
| backend/{ => pipeline}/detectors/wasb_runner.py       | 2 +-             |
| backend/pipeline/segmenters/__init__.py               | 16 ++++          |
| backend/{indexer => pipeline/segmenters}/base.py      | 2 +-             |
| backend/{indexer => pipeline/segmenters}/gemini.py    | 4 +-             |
| backend/{indexer => pipeline/segmenters}/motion.py    | 4 +-             |
| .../segmenters}/tracknet_hybrid.py                    | 4 +-             |
| .../segmenters}/wasb_hybrid.py                        | 4 +-             |
| .../segmenters}/yolo_hybrid.py                        | 2 +-             |
| backend/{ => pipeline}/stitcher/compiler.py           | 0                |
| backend/pipeline/validators/__init__.py               | 17 ++++          |
| backend/{ => pipeline}/validators/base.py             | 0                |
| backend/{ => pipeline}/validators/blur.py             | 2 +-             |
| backend/{ => pipeline}/validators/format.py           | 2 +-             |
| .../{ => pipeline}/validators/pts_continuity.py       | 2 +-             |
| backend/{ => pipeline}/validators/relevance.py        | 2 +-             |
| .../{ => pipeline}/validators/resolution_fps.py       | 2 +-             |
| backend/{ => pipeline}/validators/scene_cut.py        | 2 +-             |
| backend/validators/__init__.py                        | 17 ----          |
| docs/CODE_MAP.md                                      | 90 ++++++++----- |
| run_eval.py                                           | 2 +-             |
| run_phase3_eval.py                                    | 4 +-             |
| run_process_and_stitch.py                             | 8 +-             |
| run_regression.py                                     | 2 +-             |
| scratch/test_ai_segmenter.py                          | 4 +-             |
| tests/test_automated_provider.py                      | 2 +-             |
| tests/test_base.py                                    | 2 +-             |
| tests/test_database.py                                | 2 +-             |
| tests/test_eval_harness.py                            | 2 +-             |
| tests/test_eval_regression.py                         | 2 +-             |
| tests/test_gemini.py                                  | 2 +-             |
| tests/test_hit_detector.py                            | 4 +-             |
| tests/test_rally_gate.py                              | 4 +-             |
| tests/test_tracknet_hybrid.py                         | 2 +-             |
| tests/test_tracknet_runner.py                         | 2 +-             |
| tests/test_wasb_cache.py                              | 2 +-             |
| tests/test_yolo_hybrid.py                             | 4 +-             |
| 55 files changed, 139 insertions(+), 139 deletions(-) |                  |

### 83. assistant (2026-06-07T08:17:30.284Z)

Now I understand the resync scope. \*\*PR #40's restructure\*\* moved everything into a layered tree:

| Old path                                    | New path                                 |
|---------------------------------------------|------------------------------------------|
| `backend/indexer/`                          | `backend/pipeline/segmenters/` (renamed) |
| `backend/validators/`                       | `backend/pipeline/validators/`           |
| `backend/detectors/`, `audio/`, `stitcher/` | `backend/pipeline/*`                     |

```
| `backend/database.py` | `backend/infrastructure/database.py` |
| `backend/static/` | `backend/api/static/` |
```

The observability branch (3 commits) touches `backend/main.py` (heavy ? conflicts with restructure's import rewrites), `backend/validators/base.py` (+15 ? file was \*moved\* on master), and adds `backend/reporting/` + 2 test files (whose imports will reference old paths). Let me inspect the specifics before touching git.

#### 84. user (2026-06-07T08:17:34.454Z)

```
=== uncommitted spec.yaml change ===
=== master's new structure: key __init__ re-exports ===
--- backend/pipeline/segmenters/__init__.py ---
from backend.pipeline.segmenters.base import segmenter_registry, BasePlaySegmenter
from backend.pipeline.segmenters.motion import MotionPlaySegmenter
from backend.pipeline.segmenters.gemini import GeminiPlaySegmenter
from backend.pipeline.segmenters.yolo_hybrid import HybridYoloSegmenter
from backend.pipeline.segmenters.tracknet_hybrid import TrackNetHybridSegmenter
from backend.pipeline.segmenters.wasb_hybrid import WasbHybridSegmenter

__all__ = [
 "segmenter_registry",
 "BasePlaySegmenter",
 "MotionPlaySegmenter",
 "GeminiPlaySegmenter",
 "HybridYoloSegmenter",
 "TrackNetHybridSegmenter",
 "WasbHybridSegmenter",
]
--- backend/pipeline/validators/__init__.py ---
from backend.pipeline.validators.base import registry, VideoValidator, ValidationResult
from backend.pipeline.validators.format import FormatValidator
from backend.pipeline.validators.resolution_fps import ResolutionFPSValidator
from backend.pipeline.validators.pts_continuity import PTSContinuityValidator
from backend.pipeline.validators.scene_cut import SceneCutDensityValidator
from backend.pipeline.validators.blur import BlurValidator
from backend.pipeline.validators.relevance import RelevanceValidator
```

### Register all default validators in sequence (dependencies first)

```
registry.register(FormatValidator())
registry.register(ResolutionFPSValidator())
registry.register(PTSContinuityValidator())
registry.register(SceneCutDensityValidator())
registry.register(BlurValidator())
registry.register(RelevanceValidator())

__all__ = ["registry", "VideoValidator", "ValidationResult"]
=== does master have a backend/reporting dir already? ===
040000 tree f89f4c5b9b365cd233ec535a109037a725be5255 annotations
040000 tree a86fd3f88950fb82573ebc8dfc9749a362cea53a api
040000 tree d6e1a9438edea9bffa3e619433b4bb4144344b60 config
040000 tree 5ca4fe2ff6158fc5750ee73c85a98d5d33a97793 eval
040000 tree 367d21c458299d514db535d274c37a887df6d2b5 infrastructure
100644 blob 3f946867d995bb8880fced81ef3770b4d289a331 main.py
040000 tree 96fed0899af7433afca009e6524d8263b459c9fe pipeline
040000 tree 14ccc92556160549d814a02146bebd59b5f1bd1a providers
040000 tree 24349187eac419ddb2e8a2aa47b749caf1d71 sports
040000 tree 768dc192374573e9c1159541a4101716d3a40f64 tools
040000 tree 6bbab47d929826c1674a56cae2e446d6d7a1f37a utils
```

#### 85. user (2026-06-07T08:17:34.563Z)

No matches found

#### 86. assistant (2026-06-07T08:17:59.769Z)

Good ? and `git diff` on spec.yaml is now empty, so the working tree is clean (the session-start "M spec.yaml" snapshot was stale). Key facts:

- `backend/reporting/` has **no** `backend.\*` imports ? it can stay put as a top-level sibling; no internal import rewrites needed.
- master's `backend/` layers are: `api, config, eval, infrastructure, pipeline, providers, sports, tools, utils, annotations` ? **no** `reporting/` yet\*\*, so adding it as a sibling is clean.
- The real work is: (1) `backend/main.py` content conflict, (2) re-apply the validators/base.py `+15` at the **moved** path `backend/pipeline/validators/base.py`, (3) fix imports in the 2 new test files.

Let me confirm working-tree cleanliness and inspect the test imports, harvest imports, the validators/base change, and the main.py divergence.

#### 87. user (2026-06-07T08:18:06.162Z)

```
=== working tree clean? ===
M tests/test_reporting_harvest.py
M tests/test_validators_base.py
=== branch's change to validators/base.py (the +15 to re-apply at new path) ===
=== imports in the two new test files (branch/disk version) ===
--- test_validators_base.py ---
from backend.pipeline.validators.base import ValidationRegistry, ValidationResult,
VideoValidator
--- test_reporting_harvest.py ---
import os
import pytest
from backend.reporting import emit_indexer_report, harvest # noqa: E402
from backend.pipeline.validators.base import ValidationResult # noqa: E402
--- harvest.py imports ---
import obsreport.freely.
from __future__ import annotations
import os
from pathlib import Path
from typing import Any, Dict, List, Optional
import obsreport
--- reporting/__init__.py imports ---
from __future__ import annotations
import os
from typing import Any, Dict, List, Optional
```

#### 88. user (2026-06-07T08:18:37.786Z)

[Request interrupted by user]

#### 89. assistant (2026-06-10T16:20:46.962Z)

No response requested.

#### 90. user (2026-06-10T16:20:47.312Z)

@ "C:\Users\avidu\Projects\antigravity\_analysis\_results.md"  
I got this feedback (attached as md file) for this project. Can you take a very detailed look at this and provide a detailed report on this? Feel free to push back, provide an updated view (this feedback is a bit stale) and delve into research to validate any concrete and rational critic from the feedback.

#### 91. user (2026-06-10T16:21:37.779Z)

[Request interrupted by user]

**92. assistant (2026-06-10T18:56:49.365Z)**

No response requested.

**93. user (2026-06-10T18:56:49.899Z)**

run scratch/PR88\_FIX.md

**94. assistant (2026-06-10T18:57:01.489Z)**

Let me read that file to see what it asks for.

**95. user (2026-06-10T18:57:01.490Z)**

[Request interrupted by user]

**96. user (2026-06-10T18:57:13.992Z)**

Please run scratch/PR88\_FIX.mdv only and help me merge the PR

**97. user (2026-06-10T18:57:20.793Z)**

```
1 # Fix for PR #88 (docs PR merge conflict) ? instructions for Claude Code
2
3 **Context:** PR #88 (`docs/khelsutra-ui-alpha-plan`) conflicts with master because the
branch was
4 cut from `feat/async-job-model` (pre-squash-merge of #87) and overlaps #87 on two files
5 (`docs/CODE_MAP.md`, `docs/DEFERRED_HARDENING_PLAN.md`). Fix by recreating the branch
off
6 master and **dropping those two files from this PR** (their status-sync edits will ride
with
7 PR-2 instead).
8
9 Steps (PowerShell-safe, one per line):
10
11 ```
12 git checkout master
13 git pull
14 git checkout -b docs/khelsutra-plan-2
15 git checkout docs/khelsutra-ui-alpha-plan -- docs/UI_PROPOSAL.md docs/HOSTING_PLAN.md
docs/UI_ALPHA_EXECUTION_PLAN.md docs/mockups docs/archives/decisions/DECISIONS.md
docs/NEXT_STEPS.md
16 git commit -m "docs: KhelSutra UI alpha - proposal, hosting plan, execution plan,
ADR-009"
17 git push -u origin docs/khelsutra-plan-2
18 gh pr create --fill
19 gh pr close 88 --comment "Superseded by the recreated branch (conflict-free off
master)." --delete-branch
20 ```
21
22 Note: working-tree copies of `docs/CODE_MAP.md` / `docs/DEFERRED_HARDENING_PLAN.md` /
23 `docs/UI_ALPHA_EXECUTION_PLAN.md` may show further uncommitted "#87 merged" status edits
?
24 leave them; they get committed with PR-2.
25
```

**98. user (2026-06-10T19:00:47.651Z)**

Chapter marked: "PR #88 conflict fix + merge". Continue your current work.

**99. user (2026-06-10T19:01:02.795Z)**

=== branch + fresh status ===  
docs/khelsutra-ui-alpha-plan

```

M docs/UI_ALPHA_EXECUTION_PLAN.md
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR88_FIX.md
=== PR #88 state ===
{"baseRefName":"master","headRefName":"docs/khelsutra-ui-alpha-plan","mergeStateStatus":"DIRTY",
"mergeable":"CONFLICTING","number":88,"state":"OPEN","title":"docs/khelsutra ui alpha plan"}
=== fetch ===
From https://github.com/avidullu/badminton-highlight-indexer
- [deleted] (none) -> origin/chore/gemini-flash-latest-alias
- [deleted] (none) -> origin/chore/grok-handoff-codemap-runner
- [deleted] (none) -> origin/docs/adr-008-model-topology
- [deleted] (none) -> origin/docs/commercial-readiness-review-and-plan
- [deleted] (none) -> origin/docs/field-research-flyer
- [deleted] (none) -> origin/docs/next-steps-refresh
- [deleted] (none) -> origin/docs/next-steps-session-wrap
- [deleted] (none) -> origin/docs/truth-sync-audit
- [deleted] (none) -> origin/eval/gemini-wrapper-baseline
- [deleted] (none) -> origin/feat/canonical-gt-loader
- [deleted] (none) -> origin/feat/config-pydantic-package
- [deleted] (none) -> origin/feat/eval-human-labels
- [deleted] (none) -> origin/feat/featurizer-promotion-import-wall
- [deleted] (none) -> origin/feat/gen0-tal-harness
- [deleted] (none) -> origin/feat/report-debuggability-refinements
- [deleted] (none) -> origin/fix/api-path-validation
- [deleted] (none) -> origin/fix/appconfig-nested-get
- [deleted] (none) -> origin/fix/compiler-fstring-syntax
- [deleted] (none) -> origin/fix/detector-keying-video-id
- [deleted] (none) -> origin/fix/eval-gate-and-metric-footguns
- [deleted] (none) -> origin/fix/gemini-serial-workers
- [deleted] (none) -> origin/fix/gen-retry-backoff
- [deleted] (none) -> origin/fix/nested-config-wasb
- [deleted] (none) -> origin/fix/purity-test-skip-without-numpy
- [deleted] (none) -> origin/fix/reingest-destroy-after-confirm
- [deleted] (none) -> origin/fix/reliability-hardening
- [deleted] (none) -> origin/refactor/approved-directory-structure
- [deleted] (none) -> origin/test/close-coverage-inversion
- [deleted] (none) -> origin/test/oversize-channel-ci-coverage
- [deleted] (none) -> origin/test/oversize-channel-pin
 05bf49e..3c18cb9 master -> origin/master
=== origin/master tip ===
3c18cb9 feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep (#87)
05bf49e docs: field-research recruiter flyer (C13 interviews + Phase-2 recording rules) (#86)
747a7ad docs: NEXT_STEPS session wrap ? quality table, ship-target 0.66, corpus-growth
priorities (#85)
=== the branch's own commit (0653584) file list ===
subject=docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 + PR-1 status
sync

```

```

docs/DEFERRED_HARDENING_PLAN.md | 4 +-
docs/HOSTING_PLAN.md | 119 +++++
docs/NEXT_STEPS.md | 8 +-
docs/UI_ALPHA_EXECUTION_PLAN.md | 72 +++++
docs/UI_PROPOSAL.md | 106 +++++
docs/archives/decisions/DECISIONS.md | 40 +++++
docs/mockups/ui_alpha_mockup.html | 180 +++++
7 files changed, 525 insertions(+), 4 deletions(-)
=== branch vs master two-dot diff (what differs at tips) ===
docs/DEFERRED_HARDENING_PLAN.md | 4 +-
docs/HOSTING_PLAN.md | 119 +++++
docs/NEXT_STEPS.md | 8 +-
docs/UI_ALPHA_EXECUTION_PLAN.md | 72 +++++
docs/UI_PROPOSAL.md | 106 +++++

```

```
docs/archives/decisions/DECISIONS.md | 40 ++++++
docs/mockups/ui_alpha_mockup.html | 180 ++++++
7 files changed, 525 insertions(+), 4 deletions(-)
```

## 100. user (2026-06-10T19:02:01.207Z)

=== uncommitted PR-2 edits (to preserve via stash) ===

warning: in the working copy of 'docs/UI\_ALPHA\_EXECUTION\_PLAN.md', LF will be replaced by CRLF the next time Git touches it

```
diff --git a/docs/UI_ALPHA_EXECUTION_PLAN.md b/docs/UI_ALPHA_EXECUTION_PLAN.md
```

```
index 238a18d..8939a8c 100644
```

```
--- a/docs/UI_ALPHA_EXECUTION_PLAN.md
```

```
+++ b/docs/UI_ALPHA_EXECUTION_PLAN.md
```

```
@@ -56,9 +56,12 @@ deployment (DNS/tunnel/Pages/Access) starts only after the local end-to-end gate
```

```
- Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2 tester retention
```

## Working agreement

-Claude codes + runs the offline suite (this dev container has no GPU/torch/cv2 ? pure-logic only;

-machine-side verification = owner). Branch off `master`; one PR per item; no stacking; never push

-to master; owner reviews/merges each PR. Windows CRLF: stage only intended files.

+Claude (Cowork) designs, codes, writes tests + docs, and leaves a `scratch/PRn\_HANDOFF.md` per work item. \*\*Claude Code executes all git/PR mechanics\*\* (branch, commit, push, PR, conflict

+fixes, merges after owner review) by reading that handoff ? the owner never copy-pastes

+commands (ratified 2026-06-11 after PR-1/#88 friction; supersedes the command-block pattern).

+Cowork's sandbox cannot run git on mounts (ADR-009) and CI runs the offline suite on every PR.

+One PR per item; no stacking; never push to master directly. PowerShell note: no `&&`.

## Risks & mitigations

Box uptime (offline banner, alpha-acceptable) · Gemini 429 burst-throttle (serial workers,

=== the DEFERRED\_HARDENING hunk being DROPPED from this PR (rides with PR-2) ===

```
diff --git a/docs/DEFERRED_HARDENING_PLAN.md b/docs/DEFERRED_HARDENING_PLAN.md
```

```
index 5a937ca..910de2c 100644
```

```
--- a/docs/DEFERRED_HARDENING_PLAN.md
```

```
+++ b/docs/DEFERRED_HARDENING_PLAN.md
```

```
@@ -89,8 +89,8 @@ story rests on (? in BACKLOG).
```

## 3. Async job model (debt #16)

-> \*\*Status (2026-06-10): IMPLEMENTED ? PR-1 of the UI alpha plan

(`docs/UI\_ALPHA\_EXECUTION\_PLAN.md`),

-> awaiting owner review/merge.\*\* Shipped exactly as specified below: 202+`job\_id`, `GET /api/jobs/{id}`,

+> \*\*Status: DONE ? MERGED 2026-06-11 (PR #87\*\*, PR-1 of the UI alpha plan,

+> `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`). Shipped exactly as specified below: 202+`job\_id`, `GET /api/jobs/{id}`,

> `jobs` table at `user\_version` 2, ThreadPoolExecutor(1), startup sweep for orphaned jobs, static UI

> polls. See `scratch/PR1\_HANDOFF.md` for the review bundle.

## 101. user (2026-06-10T19:02:52.265Z)

=== 1. save dropped DEFERRED hunk for PR-2 ===

saved scratch/PR2\_deferred\_hardenening\_status.patch

=== 2. stash uncommitted PR-2 edit ===

warning: in the working copy of 'docs/UI\_ALPHA\_EXECUTION\_PLAN.md', LF will be replaced by CRLF the next time Git touches it

Saved working directory and index state On docs/khelsutra-ui-alpha-plan: PR-2: working-agreement + status edits (UI\_ALPHA\_EXECUTION\_PLAN)

warning: in the working copy of 'docs/UI\_ALPHA\_EXECUTION\_PLAN.md', LF will be replaced by CRLF the next time Git touches it

=== 3. master + pull ===

Your branch is behind 'origin/master' by 1 commit, and can be fast-forwarded.

(use "git pull" to update your local branch)

Switched to branch 'master'

Updating 05bf49e..3c18cb9

Fast-forward

```
backend/api/static/app.js | 33 ++++
backend/infrastructure/database.py | 141 ++++++
backend/main.py | 153 ++++++
docs/CODE_MAP.md | 6 +-
docs/DEFERRED_HARDENING_PLAN.md | 5 +
tests/test_api_endpoints.py | 24 +++-
tests/test_async_jobs.py | 272 ++++++
tests/test_database.py | 6 +-
8 files changed, 607 insertions(+), 33 deletions(-)
create mode 100644 tests/test_async_jobs.py
3c18cb9 feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep (#87)
=== 4. new branch ===
```

Switched to a new branch 'docs/khelsutra-plan-2'

=== 5. checkout kept paths from old branch ===

=== 6. verify staged ===

```
A docs/HOSTING_PLAN.md
M docs/NEXT_STEPS.md
A docs/UI_ALPHA_EXECUTION_PLAN.md
A docs/UI_PROPOSAL.md
M docs/archives/decisions/DECISIONS.md
A docs/mockups/ui_alpha_mockup.html
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
docs/HOSTING_PLAN.md | 119 ++++++
docs/NEXT_STEPS.md | 8 +-
docs/UI_ALPHA_EXECUTION_PLAN.md | 72 ++++++
docs/UI_PROPOSAL.md | 106 ++++++
docs/archives/decisions/DECISIONS.md | 40 ++++++
docs/mockups/ui_alpha_mockup.html | 180 ++++++
6 files changed, 523 insertions(+), 2 deletions(-)
```

## 102. user (2026-06-10T19:03:03.587Z)

```
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
```

```
index 8d437c2..b087251 100644
```

```
--- a/docs/NEXT_STEPS.md
```

```
+++ b/docs/NEXT_STEPS.md
```

```
@@ -1,6 +1,9 @@
```

```
Next steps (live) ? owned-model + multisport
```

```
-**Updated:** 2026-06-10 (end of the audit?M0?Gemini-battery session). **Authoritative roadmap:**
```

```
+++Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
```

```
+ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
```

```
+`docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87** ? next: PR-2
```

```
+upload/download). **Authoritative roadmap:**
```

```
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory
```

```
`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.
```

```
@@ -51,7 +54,8 @@ Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2
```



```

 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern
exists:
 `output/MahadevpuraSingles_highlights.mp4`). **Recruiter hand-out ready:
`docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md`**
 (script, ground rules, Phase-2 consent rules; owner fills ?/contact placeholders).
-6. **Housekeeping:** rotate the Gemini API key (was shared in chat); **rename repo off
"badminton"** (owner action,
+6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working;
confirm old-key
+ deletion in AI Studio); **rename repo off "badminton"** (owner action,
 in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery
completed.

Pointers / artifacts
diff --git a/docs/archives/decisions/DECISIONS.md b/docs/archives/decisions/DECISIONS.md
index 171475f..1429035 100644
--- a/docs/archives/decisions/DECISIONS.md
+++ b/docs/archives/decisions/DECISIONS.md
@@ -467,3 +467,43 @@ round-trip test against the collector's `curate export` format. Every ship-
gate

Panel artifacts: workflow `wf_471a9cdd-d1d` (3 grounded position papers + cross-examination),
2026-06-10.
+
+### ADR-009: KhelSutra Guru ? product identity, web-UI repo topology, local-first delivery
+
+**Date:** 2026-06-10 . **Status:** RATIFIED (owner, in-session)
+
+#### Decision
+
+1. **Product name: KhelSutra Guru**; domain **khelsutra.guru** (GoDaddy receipt: 3-yr
+ registration + Full Domain Protection, renews 2029-06). Earlier "khelstura" spellings were
+ typos.
+2. **The web product lives in a separate private repo `khelsutra`** ("the heart and brain"):
+ `web/` SPA (alpha), `control-plane/` at beta, product docs migrating in over time. The
+ backend
+ enablers (async jobs #16, chunked upload/download, identity scoping, correction endpoints)
+ stay in the indexer repo ? they are engine API features.
+3. **Local-first delivery:** every alpha feature is built and end-to-end-tested against
+ localhost before any cloud/DNS/deployment work starts (gates GL ? GD in
+ `docs/UI_ALPHA_EXECUTION_PLAN.md`).
+4. Alpha topology per `docs/HOSTING_PLAN.md`: free edge (Cloudflare Pages/Access/Tunnel) + the
+ owner's GPU box; GCP credits reserved for the beta control plane (Cloud Run/GCS).
+
+#### Rationale
+
+- Repo split: the npm/UI dependency tree stays outside the indexer's MIT/Apache/BSD licensing
+ gate; Pages auto-deploy wants a repo boundary; the engine repo's pending rename stops
+ blocking
+ product work. Named `khelsutra` (not `khelsutra-ui`) so the beta control plane doesn't force
+ a
+ third repo.
+- Local-first (owner-requested): removes cloud variables while the new job/upload surface
+ stabilizes; deployment becomes a configuration exercise against a proven build.
+- Planning docs stay canonical in the engine repo for now (single decision ledger; the copy-
+ drift
+ failure mode is documented in ADR-008's rationale) with pointer READMEs in `khelsutra`.
+
+#### Consequences
+
+- The UI alpha greenlights DEFERRED_HARDENING_PLAN **#16** as PR-1; the owner still gives an
+ explicit "go" before coding starts (CLAUDE.md owner-gating).
+- `khelsutra` is all-rights-reserved (no OSS license file); secrets never committed; Pages
+ build
+ root = `web/`.

```

+ - Alpha auth = Cloudflare Access JWT at the edge; the backend reads the authenticated email;  
+ local dev uses a `DEV\_USER\_EMAIL` override ? no auth code lives in the SPA.  
+ - Sandbox note: git metadata operations on the Cowork mount are unreliable (lock/unlink denied);  
+ `khelsutra` git init/commit/push run Windows-side (owner or Claude Code).

### 103. user (2026-06-10T19:03:27.232Z)

[docs/khelsutra-plan-2 6194f50] docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009  
6 files changed, 523 insertions(+), 2 deletions(-)  
create mode 100644 docs/HOSTING\_PLAN.md  
create mode 100644 docs/UI\_ALPHA\_EXECUTION\_PLAN.md  
create mode 100644 docs/UI\_PROPOSAL.md  
create mode 100644 docs/mockups/ui\_alpha\_mockup.html  
6194f50 docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009  
3c18cb9 feat(api): async job model (#16) - 202+job\_id, jobs table, worker, crash sweep (#87)  
branch 'docs/khelsutra-plan-2' set up to track 'origin/docs/khelsutra-plan-2'.  
To https://github.com/avidullu/badminton-highlight-indexer.git  
\* [new branch] docs/khelsutra-plan-2 -> docs/khelsutra-plan-2

### 104. user (2026-06-10T19:03:41.086Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/89>

### 105. user (2026-06-10T19:03:55.181Z)

=== close #88 + delete old branch ===  
? Closed pull request avidullu/badminton-highlight-indexer#88 (docs/khelsutra ui alpha plan)  
? Deleted branch docs/khelsutra-ui-alpha-plan  
=== branches after ===  
chore/gemini-flash-latest-alias  
docs/adr-008-model-topology  
docs/field-research-flyer  
\* docs/khelsutra-plan-2  
docs/next-steps-multisport  
docs/next-steps-refresh  
docs/next-steps-session-wrap  
docs/truth-sync-audit  
eval/gemini-wrapper-baseline  
feat/async-job-model  
feat/canonical-gt-loader  
feat/featurizer-promotion-import-wall  
feat/gen0-tal-harness  
fix/api-path-validation  
fix/detector-keying-video-id  
fix/eval-gate-and-metric-footguns  
fix/gemini-serial-workers  
fix/gen-retry-backoff  
fix/nested-config-wasb  
fix/purity-test-skip-without-numpy  
fix/reingest-destroy-after-confirm  
fix/reliability-hardening  
master  
test/close-coverage-inversion  
test/oversize-channel-pin  
=== #89 checks (initial) ===  
{ "mergeStateStatus": "UNSTABLE", "mergeable": "MERGEABLE", "statusCheckRollup": [ { "\_\_typename": "CheckRun", "completedAt": "0001-01-01T00:00:00Z", "conclusion": "", "detailsUrl": "https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27299399399/job/80640731684", "name": "Syntax sweep + import smoke (no GPU stack)", "startedAt": "2026-06-10T19:03:47Z", "status": "IN\_PROGRESS", "workflowName": "CI" }, { "\_\_typename": "CheckRun", "completedAt":

```
:"0001-01-01T00:00:00Z","conclusion":"","detailsUrl":"https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27299399399/job/80640731719","name":"Full test suite (offline, mocked)","startedAt":"2026-06-10T19:03:47Z","status":"IN_PROGRESS","workflowName":"CI"]}]}
```

#### 106. user (2026-06-10T19:05:28.811Z)

```
[00:34:11] Syntax sweep + import smoke (no GPU stack)=COMPLETED/SUCCESS; Full test suite (offline, mocked)=IN_PROGRESS/ (mergeState=UNSTABLE)
[00:34:36] Syntax sweep + import smoke (no GPU stack)=COMPLETED/SUCCESS; Full test suite (offline, mocked)=IN_PROGRESS/ (mergeState=UNSTABLE)
[00:35:02] Syntax sweep + import smoke (no GPU stack)=COMPLETED/SUCCESS; Full test suite (offline, mocked)=IN_PROGRESS/ (mergeState=UNSTABLE)
[00:35:28] Syntax sweep + import smoke (no GPU stack)=COMPLETED/SUCCESS; Full test suite (offline, mocked)=COMPLETED/SUCCESS (mergeState=CLEAN)
ALL CHECKS SUCCESS
```

#### 107. user (2026-06-10T19:05:49.656Z)

```
=== merge #89 ===
Updating 3c18cb9..34cf26c
Fast-forward
From https://github.com/avidullu/badminton-highlight-indexer
 * branch master -> FETCH_HEAD
 3c18cb9..34cf26c master -> origin/master
docs/HOSTING_PLAN.md | 119 ++++++
docs/NEXT_STEPS.md | 8 +-
docs/UI_ALPHA_EXECUTION_PLAN.md | 72 ++++++
docs/UI_PROPOSAL.md | 106 ++++++
docs/archives/decisions/DECISIONS.md | 40 ++++++
docs/mockups/ui_alpha_mockup.html | 180 ++++++
6 files changed, 523 insertions(+), 2 deletions(-)
create mode 100644 docs/HOSTING_PLAN.md
create mode 100644 docs/UI_ALPHA_EXECUTION_PLAN.md
create mode 100644 docs/UI_PROPOSAL.md
create mode 100644 docs/mockups/ui_alpha_mockup.html
=== sync master ===
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
34cf26c docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 (#89)
3c18cb9 feat(api): async job model (#16) - 202+job_id, jobs table, worker, crash sweep (#87)
=== restore PR-2 working-tree edits: stash pop ===
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
 (use "git add <file>..." to update what will be committed)
 (use "git restore <file>..." to discard changes in working directory)
 modified: docs/UI_ALPHA_EXECUTION_PLAN.md

Untracked files:
 (use "git add <file>..." to include in what will be committed)
 artifacts/
 scratch/HIGH_RISK_REMEDIATION_PLAN.md
 scratch/PR1_BODY.md
 scratch/PR1_HANDOFF.md
 scratch/PR2_deferred_hardening_status.patch
 scratch/PR88_FIX.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (d82778316dd7876e05c8ea51edcc761866b71d28)
=== re-apply dropped DEFERRED hunk for PR-2 ===
=== final status ===
master
M docs/DEFERRED_HARDENING_PLAN.md
```

```
M docs/UI_ALPHA_EXECUTION_PLAN.md
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

#### 108. user (2026-06-10T19:06:02.822Z)

```
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-10 (GEMINI BATTERY COMPLETE ? full coverage; owner's billing was
fine, postpay active; ALL
13 failures were burst/capacity artifacts). Merged this stretch: **80** (chunker re-
encode/downscale, from the chip
14 session ? reviewed+merged), **81** (rescued the chip branch's orphaned run_signals test
via cherry-pick), **83**
15 (provider: 6 generate attempts exponential backoff+jitter ? saved chunks live), **84**
(tracked
16 `eval_baselines/gemini_wrapper_2026-06-10.json`). Suite 336 green.
17 - **FINAL QUALITY TABLE (IoU .5 vs human golden labels):**
18 - **Mahadevpura** (clean-ish, 31 rallies): heuristic 0.657 · Gen-0 0.744 · **two-stage
WASB+Gemini refine 0.83**
19 (P .81 R .84, 32 preds) · **full-video wrapper 0.93** (P 1.0 R .87).
20 - **Kushagra** (multi-court, 32 rallies, FULL 7/7 coverage + audio): heuristic 0.157 ·
Gen-0 0.561 ·
21 **wrapper 0.66** (P .69 R .62, ±1.2s; FPs = background-game pickups + one 53s dead-
time merge ? exactly the
22 contrast-metric confound).
23 - **STRATEGY (refined):** clean footage commoditized (0.93 cheap); **multi-court drops
the wrapper to 0.66 = the
24 owned model's ship target** (Gen-0 is 0.10 behind at n=6 ? within corpus-growth
reach). Two-stage refine quality
25 (0.83) < full wrapper on short videos; its value = COST on long academy tapes (only
candidate clips uploaded).
26 Wrapper reliability is genuinely flaky (503 waves all session) ? provider-fallback
architecture is load-bearing.
27 - Artifacts: `output/mahadevpura_gemini_refined.csv`,
`output/MahadevpuraSingles_highlights.mp4` (demo reel),
28 Kushagra preds under the ORIGINAL corpus MD5 now. Total Gemini spend today: ~$1-2.
29 - PENDING/owner: rotate the chat-pasted key; ship-gate F1@tIoU decision (multi-court
target now has a number: 0.66);
30 C13 interviews + camera pilot; repo rename.
31
32 **Checkpoint:** 2026-06-10 (GEMINI BASELINE LANDED ? owner's "gradual ramp-up" theory
CORRECT). New key in gitignored
33 `.env` (rotate later ? pasted in chat). The "credits depleted" 429 was a BURST-THROTTLE
artifact: text ping + serial
34 upload worked ? **PR #79 merged**: gemini segmenter now honors `indexing.ai_max_workers`
(was hardcoded 5; config ships
35 1=serial ? at 5 workers the same video 429'd, at 1 it worked).
36 - ? **WRAPPER BASELINE (gemini-flash-latest, ~$0.2-0.5 total): Mahadevpura F1 0.93** (P
1.00 R 0.87, mIoU 0.80, ±0.7s;
37 3 of 4 misses = short rallies Gemini FOUND but our min_segment_duration=3.0 rejected)
```

vs Gen-0 learned 0.744 vs  
38 heuristic 0.657. **\*\*Clean footage = commoditized.\*\***  
39 - ?? **\*\*MOAT TEST** (Kushagra multi-court, windowed n=12 GT): wrapper F1 0.56\*\* (P .54 R .58 mIoU .69) ? Gen-0 learned  
40 0.561, both >> heuristic 0.157. **\*\*The multi-court confound hurts Gemini too ? the wedge survives on the target**

#### 109. user (2026-06-10T19:06:18.490Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 110. assistant (2026-06-10T19:06:34.556Z)

Done ? **\*\*[#89](https://github.com/avidullu/badminton-highlight-indexer/pull/89)** is merged to master\*\* (`34cf26c`), replacing the conflicted #88.

### What I ran (per [scratch/PR88\_FIX.md](scratch/PR88\_FIX.md))

- \*\*Verified the diagnosis first\*\***: #88 was `CONFLICTING` because its branch carried #87's pre-squash commits; the fast-forward of master confirmed #87 touched exactly `CODE\_MAP.md` + `DEFERRED\_HARDENING\_PLAN.md` ? the two files the doc drops. The kept paths differed from new master *only* by the KhelSutra edits, so the recreate was safe.
- Recreated `docs/khelsutra-plan-2` off post-#87 master, checked out the 6 kept paths, verified the staged diff was purely additive (ADR-009 append + NEXT\_STEPS status sync, no #87 reversions), committed, pushed.
- Opened **\*\*#89\*\*** (body notes it supersedes #88 and why), closed **\*\*#88\*\*** with the prescribed comment + deleted its branch (local and remote).
- \*\*Merge\*\***: waited for CI per your merge discipline (`statusCheckRollup`, not a piped watch) ? both checks (`Syntax sweep`, `Full test suite`) completed **\*\*SUCCESS\*\***, `mergeState=CLEAN` ? then squash-merged with branch delete and synced local master.

### One deviation from the doc's literal steps, for safety

`docs/UI\_ALPHA\_EXECUTION\_PLAN.md` had **\*\*uncommitted\*\*** edits (the new working-agreement text) which the doc says must ride with PR-2 ? but the literal `git checkout master` would have refused on them and step 4's checkout would have clobbered them. So I stashed before the dance and popped after the merge.

### End state ? preserved for PR-2 (intentionally uncommitted on master)

- `docs/UI\_ALPHA\_EXECUTION\_PLAN.md` ? the working-agreement rewrite (Cowork designs + handoffs; Claude Code runs git/PR mechanics).  
- `docs/DEFERRED\_HARDENING\_PLAN.md` ? the "#16 DONE ? MERGED (PR #87)" status hunk that was dropped from #89; also saved as [scratch/PR2\_deferred\_hardenening\_status.patch](scratch/PR2\_deferred\_hardenening\_status.patch) as insurance.

Don't `checkout --` those two away ? they're PR-2's payload. Checkpoint recorded in memory. Next UI-alpha slice per the execution plan is **\*\*PR-2 (upload/download)\*\*** whenever you give the go.

#### 111. user (2026-06-10T19:06:39.209Z)

can you merge the PR as well for me and make sure its clean slate now?

#### 112. assistant (2026-06-10T19:07:19.420Z)

#89 was already merged in my last turn ? let me re-verify that, check for any other open PRs (both repos), and then do the clean-slate sweep. Note the fetch earlier showed ~24 stale local branches whose remotes were deleted; I'll map each to its PR state before pruning anything.

#### 113. user (2026-06-10T19:07:26.595Z)

```

=== #89 state ===
{"mergeCommit":{"oid":"34cf26c6f80f9a6e9e6be11e7cbc0c2f8219761f"},"mergedAt":"2026-06-10T19:05:43Z","number":89,"state":"MERGED"}
=== open PRs (indexer) ===
=== fresh status + master sync ===
master
 M docs/DEFERRED_HARDENING_PLAN.md
 M docs/UI_ALPHA_EXECUTION_PLAN.md
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
From https://github.com/avidullu/badminton-highlight-indexer
- [deleted] (none) -> origin/docs/khelsutra-plan-2
- [deleted] (none) -> origin/docs/khelsutra-ui-alpha-plan
34cf26c docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 (#89)
34cf26c docs: KhelSutra UI alpha - proposal, hosting plan, execution plan, ADR-009 (#89)
=== local branches now ===
chore/gemini-flash-latest-alias
docs/adr-008-model-topology
docs/field-research-flyer
docs/next-steps-multisport
docs/next-steps-refresh
docs/next-steps-session-wrap
docs/truth-sync-audit
eval/gemini-wrapper-baseline
feat/async-job-model
feat/canonical-gt-loader
feat/featurizer-promotion-import-wall
feat/gen0-tal-harness
fix/api-path-validation
fix/detector-keying-video-id
fix/eval-gate-and-metric-footguns
fix/gemini-serial-workers
fix/gen-retry-backoff
fix/nested-config-wasb
fix/purity-test-skip-without-numpy
fix/reingest-destroy-after-confirm
fix/reliability-hardening
master
test/close-coverage-inversion
test/oversize-channel-pin

```

#### 114. user (2026-06-10T19:07:28.591Z)

```

=== open PRs (collector) ===
=== collector branch/status ===
master
 M .gitignore

```

#### 115. user (2026-06-10T19:08:11.394Z)

```

=== DELETED (23) ===
chore/gemini-flash-latest-alias (PR #78 MERGED)
docs/adr-008-model-topology (PR #73 MERGED)
docs/field-research-flyer (PR #86 MERGED)
docs/next-steps-multisport (PR #62 MERGED)
docs/next-steps-refresh (PR #72 MERGED)
docs/next-steps-session-wrap (PR #85 MERGED)
docs/truth-sync-audit (PR #71 MERGED)
eval/gemini-wrapper-baseline (PR #84 MERGED)
feat/async-job-model (PR #87 MERGED)
feat/canonical-gt-loader (PR #74 MERGED)

```

```

feat/featurizer-promotion-import-wall (PR #75 MERGED)
feat/gen0-tal-harness (PR #77 MERGED)
fix/api-path-validation (PR #68 MERGED)
fix/detector-keying-video-id (PR #64 MERGED)
fix/eval-gate-and-metric-footguns (PR #67 MERGED)
fix/gemini-serial-workers (PR #79 MERGED)
fix/gen-retry-backoff (PR #83 MERGED)
fix/nested-config-wasb (PR #66 MERGED)
fix/purity-test-skip-without-numpy (PR #76 MERGED)
fix/reingest-destroy-after-confirm (PR #65 MERGED)
fix/reliability-hardening (PR #69 MERGED)
test/close-coverage-inversion (PR #70 MERGED)
test/oversize-channel-pin (PR #81 MERGED)
=== KEPT / FLAGGED (0) ===
=== remaining branches ===
* master

```

#### 116. user (2026-06-10T19:08:22.257Z)

```

=== artifacts/ contents (indexer) ===
artifacts\rally_tal_gen0 (0KB)
artifacts\rally_tal_gen0\0.1.0 (0KB)
artifacts\rally_tal_gen0\0.1.0\manifest.json (6.4KB)
artifacts\rally_tal_gen0\0.1.0\weights.json (0.9KB)
=== collector .gitignore diff ===
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git
touches it
diff --git a/.gitignore b/.gitignore
index 52dac92..ac1420f 100644
--- a/.gitignore
+++ b/.gitignore
@@ -53,3 +53,7 @@ data/temp/

Local DB backups (e.g. data/sports_metadata.backup-YYYY-MM-DD.db) ? never commit
data/*.backup-*.db
+
+# Model artifact bundles: weights are NEVER committed (private Releases); the
+# reviewable surface is each bundle's manifest.json (ADR-008).
+artifacts/

```

#### 117. user (2026-06-10T19:08:58.107Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 118. assistant (2026-06-10T19:09:14.480Z)

The PR was already merged in my previous turn ? re-confirmed now:  
**\*\*[#89](https://github.com/avidullu/badminton-highlight-indexer/pull/89) is MERGED\*\*** (squash commit `34cf26c` on master). What this turn added is the clean-slate sweep:

#### Slate status ? indexer

- **\*\*No open PRs.\*\*** Local `master` == `origin/master` at `34cf26c` (the #89 squash).
- **\*\*All 23 stale local branches pruned.\*\*** Before deleting each one, I mapped it to its PR and confirmed the PR state was MERGED (#62?#87 ? the audit/hardening/M0/Gemini mega-session leftovers, plus #87 async jobs). Nothing was force-deleted without a merged PR backing it; only `master` remains.
- **\*\*Remote is equally clean\*\*** ? the merge auto-deleted both khelsutra branches, and the prune confirmed no stale remote refs.

#### What remains in the tree ? intentional, do not clean

- `docs/DEFERRED\_HARDENING\_PLAN.md` + `docs/UI\_ALPHA\_EXECUTION\_PLAN.md` (modified) ? the \*\*PR-2 payload\*\* preserved per [PR88\_FIX.md](scratch/PR88\_FIX.md) (also backed up as [PR2\_deferred\_hardening\_status.patch](scratch/PR2\_deferred\_hardening\_status.patch)).

- `scratch/` handoff files (untracked by convention) and `artifacts/rally\_tal\_gen0/0.1.0/` ? the Gen-0 model bundle (manifest + weights), intentional ADR-008 local state.

## One flag ? collector repo

The collector is also clean (no open PRs, on master) \*\*except\*\* one stray uncommitted `.gitignore` edit: it adds an `artifacts/` ignore rule with an ADR-008 comment ("weights never committed; manifest is the reviewable surface"). It looks like a forgotten leftover from the mega-session ? sensible content, just never committed. I left it untouched since opening unrequested PRs is against the repo convention. Say the word and I'll turn it into a micro-PR, or it can ride with the next collector change.

Memory checkpoint updated. Next up whenever you're ready: \*\*PR-2 (upload/download)\*\* from the execution plan.

### 119. user (2026-06-10T19:09:16.870Z)

Go for PR2 and if you complete it then go ahead and merge it and keep the docs updated!

### 120. assistant (2026-06-10T19:09:42.902Z)

PR-2 it is. The working agreement says implementation rides from the execution plan, so let me first read the plan and the PR-1 handoff to know exactly what PR-2 specifies.

### 121. user (2026-06-10T19:09:43.992Z)

```
1 # UI Alpha Execution Plan ? KhelSutra Guru
2
3 **Status:** ACTIVE · updated 2026-06-10 · **Linked:** `UI_PROPOSAL.md` ·
`HOSTING_PLAN.md` ·
4 `mockups/ui_alpha_mockup.html` · ADR-009 (`archives/decisions/DECISIONS.md`).
5 **Sequencing principle (owner-requested):** build and test **everything locally first**;
```

| Date       | Decision                                                                                                                                                    |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-06-10 | Product name **KhelSutra Guru** · domain **kheelsutra.guru** (GoDaddy, 3-yr + protection)                                                                   |
| 2026-06-10 | Separate **private repo `kheelsutra`** ("heart and brain"): `web/` SPA now, control plane at beta; backend enablers stay in the indexer repo                |
| 2026-06-10 | **Local-first delivery** (this plan's phase structure)                                                                                                      |
| open       | Final "go" on PR-1 coding start (= #16 greenlight) · training opt-in default OFF · invite list · stack (React+Vite+TS+Tailwind stands unless owner objects) |

```
16
17 ## Master checklist
18
19 ### G0 ? Setup (now)
20 - [x] Name/domain/repo-split ratified; spelling confirmed via GoDaddy receipt
21 - [x] `kheelsutra` repo **scaffolded** at `C:\Users\avidu\Projects\kheelsutra` (README,
.gitignore, web/, docs/)
22 - [x] **Owner:** git-init + private GitHub push ? done 2026-06-10:
`https://github.com/avidullu/kheelsutra` (private; git runs Windows-side per ADR-009)
23
24 - [x] **Owner:** explicit "go" for PR-1 ? given 2026-06-10
25 - [x] **Owner:** Gemini API key rotated ? 2026-06-10, owner-verified (confirm old-key
deletion in AI Studio)
26
27 ### Phase L ? build & test locally (no cloud; ?6?7 sessions)
28
29 **L1 ? backend enablers** (indexer repo · branch off master · ONE PR each, no stacking ·
```



```

offline tests in every PR · GPU/real-video checks = owner):
30 - [x] **PR-1 · B1 async job model (#16):** `jobs` table (user_version 1?2), `POST
/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,
restart/crash state handling ? **MERGED 2026-06-11, PR #87** (CI green: 333 passed; machine-side
real-video checks fold into L3)
31 - [] **PR-2 · B2+B3 upload & download:** chunked `POST /api/upload` (?90 MB parts ?
assemble ? MD5 ? register under `allowed_video_root/<user>/`; size cap + mp4/mov allowlist) +
streaming `GET /api/highlights/{video_id}/{name}`
32 - [] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every
query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;
rate limit; per-user quota
33 - [] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-
nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire
`backend/api/static` (B5)
34
35 **L2 ? SPA** (khelsutra repo, `web/`; dev server proxies to `http://localhost:8000`):
36 - [] **M1 scaffold:** Vite + React + TS + Tailwind, router, typed API client, branding
(KhelSutra Guru)
37 - [] **M2:** A2 Upload (chunked, guidelines checklist) + A3 Job Progress (poll)
38 - [] **M3:** A1 Matches + A4 Rally Review (player seek, accept/reject/nudge wired to
PR-4)
39 - [] **M4:** A5 Highlights (compile + download) + A6 Report + engine-offline banner
40 - [] Unit tests for API client + review-state logic; `npm run build` clean
41
42 **L3 ? local end-to-end (owner's box, real match videos):**
43 - [] Full UI flow on localhost: upload ? job ? review/correct ? compile ? download
44 - [] `python run_all_tests.py` green; UI polling vs SQLite WAL sanity; WASB/CLI paths
untouched
45 - [] **GATE GL:** owner signs off local E2E ? only then deployment work begins
46
47 ### Phase D ? deployment (?2 sessions + ~1 hr owner)
48 - [] **D1 (owner):** Cloudflare account ? add khelsutra.guru ? GoDaddy nameserver
switch (HOSTING_PLAN $5) ? scoped API token ($6) ? Pages project connected to GitHub `khelsutra`
(build root `web/`)
49 - [] **D2 (Claude):** DNS records; `cloudflared` Windows service on the box ?
`api.khelsutra.guru`; Access apps + email-allowlist policies for `app` & `api`; SPA prod env
50 - [] **D3:** deployed smoke E2E (owner-only allowlist); go-live security checklist
(HOSTING_PLAN $7) all green
51 - [] **GATE GD:** checklist signed off
52
53 ### Phase R ? rollout
54 - [] Wave 1: 5 testers (C13/camera-pilot contacts) + feedback channel (WhatsApp group
or Google Form)
55 - [] 2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL $4)
56 - Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2
tester retention
57
58 ## Working agreement
59 Claude (Cowork) designs, codes, writes tests + docs, and leaves a
`scratch/PRn_HANDOFF.md` per
60 work item. **Claude Code executes all git/PR mechanics** (branch, commit, push, PR,
conflict
61 fixes, merges after owner review) by reading that handoff ? the owner never copy-pastes
62 commands (ratified 2026-06-11 after PR-1/#88 friction; supersedes the command-block
pattern).
63 Cowork's sandbox cannot run git on mounts (ADR-009) and CI runs the offline suite on
every PR.
64 One PR per item; no stacking; never push to master directly. PowerShell note: no `&&`.
65
66 ## Risks & mitigations
67 Box uptime (offline banner, alpha-acceptable) · Gemini 429 burst-throttle (serial
workers,
68 `ai_max_workers=1`, 6-attempt backoff ? learned 2026-06-10) · CF free-tier 100 MB body
cap
69 (chunked upload design) · single-GPU box (alpha default = Gemini path; WASB jobs queue-

```

```

depth 1) .
70 .guru renewal cost ?18.8k/3-yr (optional: transfer to Cloudflare Registrar at-cost after
the
71 60-day registration lock; domain-protection may need toggling for transfer).
72
73 ## Session map (default order)
74 S1?S2 PR-1 . S3 PR-2 . S4 PR-3 . S5 PR-4 . S6 M1+M2 . S7 M3 . S8 M4 . S9 L3 E2E + fixes
.
75 S10 D1?D3 . S11 wave-1 launch
76

```

## 122. user (2026-06-10T19:09:44.218Z)

```

1 # PR-1 handoff ? async job model (#16) . code complete, awaiting your review
2
3 **Date:** 2026-06-10 (overnight build, per your "go") . **Spec:**
`docs/DEFERRED_HARDENING_PLAN.md` §3
4 . **Plan:** `docs/UI_ALPHA_EXECUTION_PLAN.md` Phase L ? PR-1 . **Built by:** Claude
(Cowork session)
5
6 ## ?? One thing before review: run the suite
7
8 The Cowork sandbox's Linux bridge failed mid-session (a bad WSL-path mount wedged it ?
every
9 shell call now errors), so the test suite is **written but was NOT executed by me**.
First step
10 on your machine (or hand to Claude Code):
11
12 ```
13 python run_all_tests.py # full offline suite
14 python -m pytest tests/test_async_jobs.py -v # just the new contract tests (11 tests)
15 ```
16
17 If anything is red, don't debug at midnight ? leave it and I'll fix in the next session.
18
19 ## What changed (PR-1 file list ? exactly these 7 for the branch)
20
21 | File | Change |
22 |---|---|
23 | `backend/infrastructure/database.py` | Migration `user_version` 1?2: `jobs` table +
`idx_jobs_status`; job store methods (`create_job`, `mark_job_running`, `set_job_progress`,
`set_job_video_id`, `mark_job_done`, `mark_job_failed`, `get_job`, `fail_stale_jobs`) |
24 | `backend/main.py` | `POST /api/process` ? fast-fail 4xx then **202 + job_id**; new
`GET /api/jobs/{job_id}`; `_execute_processing` (former sync body, behavior-preserving) +
`_run_job` worker + lazy `ThreadPoolExecutor(max_workers=1)`; startup sweep `fail_stale_jobs()`
in `main()` |
25 | `backend/api/static/app.js` | Submit handler polls `/api/jobs/{id}` every 3 s with
stage display (also fixes the pre-existing `data.rallies` ? `data.play_segments` field bug) |
26 | `tests/test_async_jobs.py` | **New** ? 11 tests: migration (fresh v2 + v1?v2 upgrade),
submit 202, fast-fail 400/404, full lifecycle, pipeline-verdict-vs-job-status, crash restore,
progress stages, unknown job 404, stale sweep, real-executor cross-thread run |
27 | `tests/test_api_endpoints.py` | Process tests updated to the 202+poll contract (inline
executor; query/compile/config untouched) |
28 | `docs/DEFERRED_HARDENING_PLAN.md` | §3 status banner: implemented, awaiting review |
29 | `docs/CODE_MAP.md` | §2.0 main.py (async surface), §2.7 DB (v2 + job store), §3 jobs-
row contract |
30
31 ## Design decisions you're reviewing
32
33 1. **Job `status` = EXECUTION state** (`queued|running|done|failed`). A video that fails
34 *validation* is a job that finished fine: `status="done"`,
`result["status"]="failed"`.
35 `failed` means the worker crashed/raised. (Keeps "did the machinery work" separable
from
36 "what did the pipeline conclude" ? the UI report screen needs both.)

```

```

37 2. **`max_workers=1`** ? single GPU box serializes work, and the Gemini provider is
38 rate-fragile under parallel uploads (your 2026-06-10 ops learning). Segmenters still
39 parallelize AI handoff internally. Knob can move to config later if ever needed.
40 3. **Fast-fail at submit** ? bad path / missing file / unknown segmenter still 4xx
41 immediately; everything slow (including MD5 hashing, which is minutes on 4 GB files)
42 moved into the worker.
43 4. **Raise-path now restores pre-run rows** ? previously a *raising* (vs
44 `Failure`-returning)
45 segmenter could leave half-written rows; the worker path wraps it with the same
46 destroy-AFTER-confirm prune the `Failure` branch uses. Tested.
47 5. **CLI mode untouched** ? `python backend/main.py --video-path ...` still runs
48 synchronously
49 (it's not the API; blocking is correct UX there).
50 6. **Report contract preserved** ? `emit_indexer_report` still runs in `finally:`, paths
51 still
52 grafted into the response, which now lives in `result` (the report is the job's
53 artifact,
54 per the plan).
55
56 ## Branch & commit (Windows-side, since git can't run through my mount)
57
58 ```
59 cd C:\Users\avidu\Projects\badminton-highlight-indexer
60 git checkout -b feat/async-job-model
61 git add backend/infrastructure/database.py backend/main.py backend/api/static/app.js
62 tests/test_async_jobs.py tests/test_api_endpoints.py docs/DEFERRED_HARDENING_PLAN.md
63 docs/CODE_MAP.md
64 git commit -m "feat(api): async job model - 202+job_id, jobs table (uv2), worker, crash
65 sweep (#16)"
66 ```
67
68 **Separate docs commit/PR** (planning docs are their own work item ? keeps PR-1
69 reviewable):
70
71 ```
72 git checkout master
73 git checkout -b docs/khelsutra-ui-alpha-plan
74 git add docs/UI_PROPOSAL.md docs/HOSTING_PLAN.md docs/UI_ALPHA_EXECUTION_PLAN.md
75 docs/mockups/ui_alpha_mockup.html docs/archives/decisions/DECISIONS.md docs/NEXT_STEPS.md
76 git commit -m "docs: KhelSutra Guru UI alpha - proposal, hosting plan, execution plan,
77 ADR-009"
78 ```
79
80 (`scratch/PR1_HANDOFF.md` itself: leave untracked or delete after merge.)
81
82 ## Suggested PR description (paste-ready)
83
84 > **feat(api): async job model (#16) ? 202+job_id, jobs table, worker, crash sweep**
85 >
86 > Implements DEFERRED_HARDENING_PLAN §3 exactly: `POST /api/process` returns
87 202+`job_id`
88 > with submit-time fast-fail; `GET /api/jobs/{id}` exposes `{status, progress, result,
89 > reports}`; pipeline runs on a `ThreadPoolExecutor(1)` worker; job state persists in a
90 > `jobs` table (`user_version` 2); startup sweeps orphaned jobs to `failed`; static UI
91 polls.
92 > Job status = execution state; pipeline verdict stays in `result.status`. Raising
93 segmenters
94 > now restore pre-run segments (destroy-after-confirm). CLI path unchanged.
95 > First slice of the KhelSutra UI alpha (`docs/UI_ALPHA_EXECUTION_PLAN.md`, PR-1).
96
97 ## Machine-side verification checklist (the parts only your box can test)
98
99 - [] `python run_all_tests.py` ? green (see warning at top ? not yet executed anywhere)
100 - [] Server mode: `python -m backend.main --server` ? POST a real video ? 202 ? poll
101 `/api/jobs/{id}` ? `done` with same segment results as the old sync path

```

```
89 - [] Report files written next to the video as before; paths visible in
`result.reports`
90 - [] Kill the server mid-job ? restart ? job shows `failed` with "interrupted by server
91 restart" (startup sweep) ? spec's explicit verification case
92 - [] Static UI demo: submit via the browser, watch the stage label tick, rally list
renders
93 (the `play_segments` fix means rallies actually appear now ? they never did
before)
94 - [] CLI regression: `python -m backend.main --video-path <short.mp4>` unchanged
95
```

**123. user (2026-06-10T19:09:55.861Z)**

[Request interrupted by user]